

Protecting Workers, Replacing Workers? Inclusive Labor Policies and Automation Adoption

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Abstract

This paper investigates how migrant-inclusive labor policies affect firms' technology choices. We construct a novel, multidimensional city-year inclusiveness index and combine it with firm-level panel data. Using a dialect-network instrumental variable strategy, we find that migrant inclusion policies significantly accelerate robot adoption. Decomposing the policy index reveals that the compliance-cost channel drives this automation response: provisions expanding public-service access and labor-rights protection raise the cost and rigidity of formal employment, pushing firms toward labor-saving capital. Conversely, administrative burden relief acts as a mitigating factor. Furthermore, we find that this capital-labor substitution is concentrated among firms with high labor intensity, limited financial capacity, and strict legal enforcement. Our findings demonstrate that when inclusive policies transform flexible migrant labor into heavily regulated employment, firms substitute machines for labor, highlighting a critical trade-off between labor formalization and workforce displacement.

1 Introduction

When governments strengthen labor protections, what determines whether firms adapt by formalizing their workforce or by replacing human workers with machines? A large literature shows that automation responds to changes in the price and availability of labor, including minimum-wage increases, demographic scarcity, and immigration-driven labor-supply shocks [Fan et al., 2021, Acemoglu and Restrepo, 2022, Mann and Pozzoli, 2024]. Yet wages and labor supply capture only part of the economic cost of employment. Beyond direct payroll, firms must bear social-insurance obligations, employment-protection costs, adjustment frictions, and legal risks, costs that are fundamentally shaped by labor-market institutions. This paper examines how these institutional components of labor costs shape firms' capital-labor choices.

We study this missing margin through migrant inclusive labor policies, which extend migrant workers' access to public service, social protection, and enforceable labor rights. These policies change not merely how many workers are available to firms, but the strict terms on which they are employed. The net impact on automation, however, is theoretically ambiguous due to two opposing channels. On the one hand, by granting migrants better access to residence status, public services, social security, and employment support, cities may reduce mobility frictions, strengthen workers' attachment to destination cities, and stabilize recruitment. This labor-supply expansion channel would make labor-intensive production more viable and weaken firms' incentives to substitute machines for workers. On the other hand, by improving migrants' outside options, extending social-insurance obligations, strengthening labor-rights enforcement, and making employment relationships more visible to regulators and courts, the same policies may raise the cost and rigidity of labor [Besley and Burgess, 2004, Autor et al., 2007]. This labor-cost and compliance channel would push firms toward labor-saving capital. Since migrant-inclusive labor policies are broader institutional bundles rather than a single wage mandate, determining which channel dominates therefore requires direct quantitative evidence.

However, empirically identifying this effect poses two challenges. First, migrant-inclusive labor policies are inherently difficult to quantify. Unlike minimum-wage changes, which provide a clearly observed numerical treatment, migrant inclusion is implemented through heterogeneous policy provisions covering residence eligibility, public-service access, social security, labor-rights enforcement, employment support, and administrative procedures. These components may have different, and even opposing implications for firms. Therefore, a single reform dummy or broad city characteristics cannot capture the relevant policy variation. Second, empirically testing this question requires a setting where firms face a viable trade-off between migrant labor and automation. For migrant-inclusive policies to constitute an institutional shock, migrant workers must have previously faced substantial barriers to local welfare and labor protection. Similarly, capital substitution requires industrial robots to be sufficiently accessible to serve as a realistic

production alternative. Furthermore, connecting these two factors requires the manufacturing sector to maintain a heavy reliance on migrant labor. In the absence of these conditions, it would be difficult to observe whether migrant-inclusive labor policies induce firms to automate.

China provides an ideal case because it combines all three required conditions: strong institutional barriers for migrants, large-scale reliance on migrant labor, and rapid robot adoption. For decades, rural migrants supplied labor to cities while remaining largely outside urban welfare and regulatory arrangements [Afridi et al., 2015, An et al., 2024]. Recent municipal reforms have shifted urban governance toward residence-based integration, expanding public-service access, social-security and welfare coordination, labor-rights protection, employment and entrepreneurship support, and administrative burden relief for migrants. This transformation makes migrant inclusion an institutional shock: it not only reduces exclusion but also formalizes employment relationships that had long operated in an institutional gray zone, making firms more exposed to social-insurance, legal enforcement, and worker-retention obligations. At the same time, China has become the world’s largest market for industrial robots [Cheng et al., 2019], making automation a feasible production response, while migrant workers remain central to manufacturing employment. The coexistence of migrant-labor dependence, institutional inclusion, rising labor costs, and high robot availability makes China a uniquely suitable setting for studying whether migrant-inclusive labor policies accelerate automation.

To bridge these research gaps, we construct a novel city-year measure of migrant-inclusive labor policies using 3,408 local policy records assigned to cities during 2014–2023. These records include policies issued at the provincial, municipal, and district or county levels, which are allocated to the relevant cities according to their geographic scope. Using these records, we construct an aggregate inclusiveness index and six separately scored dimensions: hukou and residence eligibility, social security and welfare coordination, public-service access, labor-rights protection, employment and entrepreneurship support, and administrative burden relief. This measurement strategy allows us to move beyond simple keyword counts or single reform indicators. It captures both the overall intensity of migrant inclusion and the institutional content of different policy components. We merge this city-level policy panel, lagged by one year, with a 2015–2024 panel of Chinese listed manufacturing firms, firm financial and governance data, city-level controls, and industry-year industrial robot installation data from the International Federation of Robotics. Our main outcome is a firm-level industrial robot adoption exposure measure that maps national industry-level robot installations to firms using pre-sample employment structures. This measure draws on the industry-level robot measurement and predetermined employment exposure approaches in Graetz and Michaels [2018] and Acemoglu and Restrepo [2020].

Based on this novel measurement, we estimate the impact of urban inclusive policies on corporate robot adoption. We estimate firm-level panel models with firm and year fixed effects, and further use a dialect-network instrumental-variable strategy that exploits policy diffusion

among historically linguistically connected cities. The results show that migrant-inclusive labor policies significantly raise industrial robot adoption. This automation response is notable in our IV estimates: a one-standard-deviation increase in the migrant inclusiveness index raises robot adoption by about 1.60 units, equivalent to approximately 7.07% of the sample mean. Taken together, our findings reveal a broader consequence of migrant integration. The same policies that expand migrants' access to urban services, settlement opportunities, and labor protection also change firms' incentives to rely on labor-saving capital.

To understand what drives this adjustment, we decompose the aggregate index into its six constituent dimensions, documenting a clear labor-cost and compliance channel. The upward automation response is mainly driven by provisions that expand public-service access, alongside marginal contributions from enhanced labor-rights protection and social-security coordination. As they are the core welfare policies that improve migrants' outside options and settlement intentions, they inherently increase firms' exposure to formal employment obligations. We show that following these policy expansions, firms report increases in total labor costs, per-worker wages, and social-insurance contributions. By contrast, administrative burden relief significantly reduces robot adoption. This opposing effect is informative: when policy lowers compliance friction, firms face less pressure to replace workers with machines.

Consistent with this cost-compliance interpretation, the shift towards automation is highly concentrated among firms most vulnerable to formalization pressures. The response is stronger for labor-intensive manufacturers, financially constrained firms lacking the liquidity to absorb recurring expenses, and those operating in provinces with strict legal enforcement. Together, these mechanisms indicate that firms adopt robots when inclusive policies make human labor more formal, more expensive, and harder to adjust.

This paper makes several contributions. First, this paper contributes to the emerging literature on migration and automation by showing that the relationship between migrant labor and robots hinges crucially on the institutional conditions of employment. Recent evidence shows that larger inflows of migrant workers can reduce robot adoption by expanding the supply of low-skilled routine labor [Mann and Pozzoli, 2024]. Our paper studies a different margin. We ask what happens when migrant workers become more integrated into local welfare systems and formal labor protection. In China, migrant-inclusive labor policies make migrant labor more protected, more visible to regulators, and more costly to adjust. We show that under these conditions, firms respond by adopting more industrial robots. This finding complements the labor-supply view of immigration and automation. Migrant labor may reduce automation when it increases the availability of low-cost routine workers, but it may accelerate automation when institutions raise the effective cost of employing those workers.

Second, the paper provides a more fine-grained, quantitative approach to measuring migrant-inclusive labor policies. Existing quantitative studies of migrant inclusion often evaluate specific

reforms or broader changes in local fiscal and migration policies [Sieg et al., 2023, An et al., 2024, Bo and Wang, 2025]. These measures are useful for describing broad institutional change, but they are less able to distinguish which specific policy components affect firms’ labor costs and compliance exposure. We construct a multidimensional annual city-level policy index from coded local policy documents. The index separates migrant inclusion into six concrete policy dimensions: hukou and residence eligibility, social-security and welfare coordination, public-service access, labor-rights protection, employment and entrepreneurship support, and administrative burden relief. This more specific measurement captures not only the overall evolution of migrant inclusiveness but also the distinct economic implications of different policy components. The decomposition shows that migrant inclusion is not a uniform policy shock. Public-service access and labor-rights protection act as automation-inducing push factors, whereas administrative burden relief works as a mitigating factor.

Third, the paper speaks to a broader policy tension in an era when migration policy, labor inclusion, and automation advance simultaneously. While existing evidence suggests that increasing the supply of migrant labor can reduce firms’ incentives to automate by easing low-skill labor shortages [Mann and Pozzoli, 2024], our findings show that integrating migrant workers into welfare and labor-protection systems may have the opposite effect when the costs of inclusion are concentrated on employers. The distinction between attracting migrant labor and formalizing migrant employment is therefore central to policy design. A more coherent policy design should combine migrant inclusion with complementary measures that reduce administrative friction, improve social-insurance portability, and help firms adjust without turning worker protection into a cost shock that triggers automation.

2 Institutional Background

2.1 Migrant Labor and Demographic Transition

Since the initiation of economic reforms in the late 1970s, China’s rapid urbanization has been heavily powered by large-scale rural-urban migration. However, this demographic shift operated under a unique institutional arrangement that separated labor mobility from full social membership in destination cities. The *Hukou* system, which strictly classifies citizens by their place of origin and rural or urban status, allowed rural residents to move to urban areas and supply labor to expanding manufacturing and service sectors, while access to locally funded education, health care, housing assistance and social protection remained closely tied to local household registrations. Migrants could therefore participate in urban production without being fully incorporated into the urban welfare system. This arrangement shaped a specific low-welfare mobility model in China: destination cities and firms benefited from a large and relatively flexible labor force, while

a substantial share of the costs associated with family reproduction, care, and social protection continued to be borne by migrant households and their places of origin [Chan and Zhang, 1999].

The economic consequences of this institutional separation extended beyond distributional inequality. By restricting migrants' access to local residency rights and public services, the *hukou* system affected migration decisions, constrained the spatial reallocation of labor, and limited the realization of urban agglomeration economies [Zhao, 1999, Au and Henderson, 2006]. Existing studies further suggest that internal migration restrictions generated substantial welfare and productivity losses, leaving sizable gains from more efficient spatial allocation unrealized [Whalley and Zhang, 2007, Tombe and Zhu, 2019]. In this sense, the traditional migrant labor regime was not merely a population registry; it served as a crucial pillar of China's growth model, shaping both the supply of migrant labor and the institutional conditions under which firms employed it.

For decades, the macroeconomic viability of this low-cost labor model was sustained by its sheer scale. However, this demographic foundation has recently reached a turning point. While the floating population reached a staggering 375.8 million in 2020 [National Bureau of Statistics of China, 2021], and the number of migrant workers stood at nearly 300 million in 2024 [National Bureau of Statistics of China, 2025b], the era of indefinitely elastic young labor has ended. The average migrant worker was 43.2 years old in 2024, with those aged fifty or above accounting for more than 30 percent of the migrant workforce [National Bureau of Statistics of China, 2025a]. As demographic pressures intensified and migrant workers increasingly demanded stable, family-based urban settlement, the traditional logic of treating migrants purely as a temporary, low-cost input became fundamentally unsustainable, leading to the subsequent institutional shift in urban governance.

2.2 Evolution of Urban Labor Inclusive Policies

China's shift toward migrant inclusion did not occur through a single dismantling of the *hukou* system. Instead, it unfolded gradually. Early reforms focused mainly on lowering the formal barriers to obtaining local *hukou*, especially in small and medium-sized cities. These changes made it easier for some migrants to acquire local residency, but legal eligibility for settlement did not automatically guarantee meaningful access to urban life. Migrant families continued to face barriers in school enrollment, social insurance, health care, housing assistance, and other locally provided services. The central policy question therefore broadened from whether migrants could obtain local *hukou* to whether they could live, work, and raise their families in destination cities on terms comparable to those of local residents. The 2014 *hukou* reform constituted a substantial relaxation of internal-migration restrictions and generated meaningful adjustments in local labor markets for both incumbent migrants and native workers [An et al., 2024].

The National New-Type Urbanization Plan marked an important turning point in this pro-

cess [State Council of China, 2014]. Rather than treating rural migrants simply as temporary labor, the plan emphasized their gradual incorporation into urban society. The subsequent residence-permit system provided a channel through which migrants without local hukou could gain access to selected public services and social benefits. The central government further encouraged cities to relax settlement restrictions and promote the urban integration of non-hukou residents [National Development and Reform Commission of China, 2016]. In practice, municipal governments translated these broad objectives into a diverse set of local measures. Some policies relaxed hukou or residence-permit eligibility; others improved social-insurance coordination, expanded access to education and other public services, strengthened labor-rights protection, provided employment and entrepreneurship support, or simplified registration and administrative procedures [Tian, 2024]. Migrant inclusion therefore evolved into a multidimensional policy package rather than a narrow change in household-registration rules.

These reforms produced visible improvements in migrants' access to urban welfare. By 2023, the overwhelming majority of children accompanying migrant parents were enrolled in compulsory education, substantially narrowing the gap in access to basic schooling. Migrant participation in work-injury insurance, medical insurance, and housing provident funds also expanded over time. More broadly, these changes reduced the long-standing separation between migrants' contribution to urban production and their access to urban social citizenship. They also made long-term, family-based settlement more feasible, potentially reducing worker turnover and allowing firms to recruit and retain migrant employees more easily.

From the perspective of employers, however, this transition had a second and potentially opposing consequence. Migrant inclusion did not merely increase workers' access to benefits; it also brought employment relationships that had often remained informal or weakly regulated into a more formal institutional framework. Even when public services were financed primarily by local governments, eligibility was frequently verified through employment contracts, residence records, or social-insurance contributions. Stronger labor-rights protection also increased the enforceability of wage, benefit, and employment-protection obligations. As a result, inclusive policies could raise firms' effective labor costs by increasing payroll obligations, administrative compliance, legal exposure, and the cost of adjusting employment. Previous evidence that hukou reforms affected firm-level employment adjustment is consistent with the idea that changes in migrants' institutional status also alter employers' labor decisions [Wang et al., 2021].

Moreover, not every component of migrant inclusion should affect firms in the same way. Expanded public-service access and stronger labor-rights protection may improve workers' welfare and attachment to destination cities, but they can also make employment more formal, durable, and visible to regulators. These dimensions may therefore increase the relative cost of labor and encourage firms to substitute capital for routine labor. By contrast, policies that simplify registration, documentation, and inspection procedures can reduce the administrative cost of

employing migrant workers. Such administrative burden relief may act as a mitigating factor, allowing cities to expand migrant protection without creating the same pressure for labor-saving substitution. This distinction anticipates the central empirical argument of the paper: the effect of migrant inclusion on automation depends not only on how inclusive a policy is, but also on the specific institutional obligations and administrative frictions it creates.

Finally, this policy transition was highly uneven across cities and over time. Although the central government established the broad direction of reform, municipal governments retained considerable discretion over settlement thresholds, benefit eligibility, enforcement intensity, and administrative procedures. Local policy choices consequently differed with fiscal capacity, population pressures, labor demand, and industrial structure [Zhang et al., 2019]. Existing research shows that these differences affect both firms’ employment adjustments and migrants’ occupational choices [Wang et al., 2021, Huang and Ke, 2024]. This decentralized implementation provides the institutional variation needed for our empirical analysis. It also makes a single reform dummy inadequate: capturing the evolution of migrant inclusion requires a city-year measure that distinguishes its different policy components. We therefore construct a multidimensional Urban Inclusiveness Index in the following section.

2.3 Construction of Urban Inclusiveness Index

Quantifying local migrant-inclusive policy is empirically challenging because relevant reforms lack straightforward numerical measurements and vary across institutional domains. To address this, we collect 3,408 policy documents issued between 2014 and 2023 from the PKULaw local-policy database. These documents originate from provincial, municipal, and county levels, and are mapped to specific cities based on their stated scope. We then evaluate each document based on its provisions across six key institutional domains: (1) hukou and residence eligibility, (2) social-security and welfare coordination, (3) public-service access, (4) labor-rights protection, (5) employment and entrepreneurship support, and (6) administrative burden relief.

We use this multidimensional framework to capture the institutional tension inherent in migrant integration. For instance, a residence-permit policy may simultaneously broaden access to education while imposing strict administrative registration and monitoring. By directly evaluating provisions and allowing a single document to receive non-zero scores across multiple domains, our measure distinguishes whether a specific policy merely expands public welfare or simultaneously intensifies employment compliance pressures for local enterprises.

Following the text-scoring methodologies of [Besley and Burgess, 2004, Tian, 2024], we calculate migrant-inclusive index for each document. The document-level scoring process integrates historical policy evaluations with AI-assisted classification and manual review. When full texts are unavailable, we score provisions based on their titles alone. Purely administrative notices receive

zero scores. Each dimension is evaluated independently on a scale from -2 to 2 . For the first five welfare and rights dimensions, a positive score represents a relaxation of institutional access or an improvement in social protection levels, while a negative score implies exclusion or restriction. For the sixth dimension (administrative control), a positive score corresponds to a reduction in administrative burdens, such as registration and inspection requirements. Finally, we aggregate these document-level scores to construct the city-level migrant-inclusiveness index. We report detailed construction of inclusiveness index in Appendix A.

Figure 1 plots the cumulative city-level policy score over 2014–2023 and provides a spatial overview of the underlying policy variation. The distribution of migrant-inclusive index is geographically uneven. High cumulative scores are predominantly concentrated across the southeastern seaboard, central regions, and southwestern hubs, with an additional cluster in the northeast. Lower scores are more common in western cities, while substantial variation also appears within provinces.

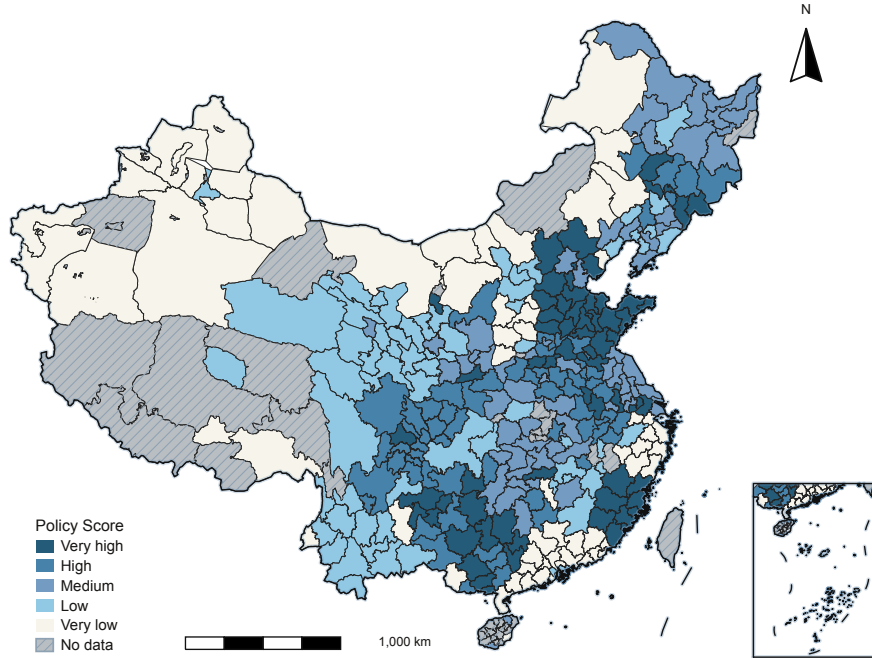


Figure 1: Spatial distribution of cumulative migrant-inclusive policy scores, 2014–2023

3 Data and Empirical Strategy

3.1 Data

Urban Inclusiveness Index(*Inclusiveness*). With the multidimensional coding framework established in Section 2.3, we construct our baseline explanatory variable by aggregating the document-level evaluations to the city-year level. Let $\mathcal{P}(c, t)$ denote the set of policy records assigned to

city c and policy year t , and let $Score_p \in \{-2, -1, 0, 1, 2\}$ denote the document-level aggregate inclusiveness score. The annual aggregate policy index is defined as:

$$PolicyIndex_{ct} = \sum_{p \in \mathcal{P}(c,t)} Score_p. \quad (1)$$

City-years without relevant policies receive zero in all indices. In our baseline specification, we lag the city-level policy index by one year before merging it with the 2015–2024 firm-level data. This one-period lag allows for policy implementation frictions and separates policy issuance from firm investment decisions. We conduct robustness checks by incorporating longer lag windows in Appendix C.

Industrial Robot Adoption(*Install*). To measure firms’ response to migrant-inclusive labor policies, we use firm-level industrial robot adoption, denoted by $Install_{ict}$. Because industrial robot installations are observed at the industry-year level rather than directly at the firm-year level, we follow the exposure logic established in [Acemoglu and Restrepo \[2020\]](#) to construct a Bartik-style exposure measure that maps national industry-level robot shocks to firms using pre-sample employment structures.

The industry shock is measured as the number of industrial robots installed in China in industry j and year t (from the IFR), scaled by baseline industry employment in 2010 (from the China Industry Statistical Yearbook). The firm-specific exposure weight is the firm’s production-staff share in 2011 (from Wind), normalized by the cross-firm median of the same share. Formally, for firm i operating in industry $j(i)$, the measure is constructed as:

$$Install_{ict} = \frac{Install_{j(i),t}^{CHN}}{Employment_{j(i),2010}} \times \frac{ProductionShare_{i,2011}}{\text{median}_k (ProductionShare_{k,2011})}. \quad (2)$$

The industry component follows the practice of measuring robot intensity using IFR industry data normalized by pre-period labor input [[Graetz and Michaels, 2018](#)]. Our firm-level adaptation uses predetermined production-staff intensity to capture differences in firms’ exposure to the same industry-year change in robot installations.

Control Variables. Our main analysis focuses on firm robot adoption and lagged city-level migrant inclusive policies. The baseline specification includes firm characteristics capturing scale and maturity, balance-sheet and operating conditions, ownership, and corporate governance. We additionally control for city per-capita GDP and science expenditure. Firm variables are obtained from CSMAR and Wind, and city variables from local statistical yearbooks. Appendix B.2 reports the complete variable list, definitions, timing conventions, and sources.

Appendix Table C1 reports the summary statistics for the main variables used in our empirical analysis. Panel A presents industrial robot adoption and the aggregate policy variable. The average value of industrial robot adoption is 22.83 units; however, the distribution exhibits a long right tail,

with a median of 8.96 and a maximum of 157.78. To mitigate the influence of extreme outliers, we winsorize the dependent variable and the main continuous policy variables at the 2nd and 98th percentiles for the baseline analysis.

Panel B reports the six decomposed policy dimensions. The distributional statistics here explicitly capture the institutional asymmetries discussed in Section 2. For instance, the dimensions capturing welfare and rights expansions, such as labor-rights protection (mean =6.40) and public-service access (mean =2.54), are strictly positive on average. In contrast, administrative burden relief has a negative mean (−1.67), indicating that inclusive provisions and migrant-specific administrative controls coexist in the policy sample.

Panels C and D summarize the firm- and city-level covariates used in the baseline analysis. State-owned enterprises account for approximately 37% of the firm-year observations. Panel E reports the labor-cost, social-protection, and employment outcomes used to examine the proposed employment-cost channel and subsequent workforce adjustment. Panel F presents the three predetermined firm characteristics central to the main heterogeneity analysis—pre-sample labor intensity, the mean FCP score during 2010–2014, and local legal environment—together with the additional grouping variables used in the supplementary analyses. Appendix B provides complete definitions and units.

3.2 Econometric Model Specification

In the baseline specification, we investigate how lagged migrant integration policy in city c affects firm i 's robot adoption. We estimate the following two-way fixed effects model:

$$\text{Install}_{ict} = \alpha + \beta \text{Inclusiveness}_{c,t-1} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \mathbf{W}'_{ct}\boldsymbol{\delta} + \mu_i + \tau_t + \varepsilon_{ict}, \quad (3)$$

The dependent variable, Install_{ict} , measures industrial robot adoption by firm i located in city c in year t . The main explanatory variable, $\text{Inclusiveness}_{c,t-1}$, is the migrant-inclusiveness index for city c , lagged by one year. This timing allows policy provisions to take effect before firms adjust their production and investment decisions and establishes a clear temporal ordering between policy measurement and the outcome. It also reduces the concern that contemporaneous industrial robot adoption mechanically affects the policy index.

We include a rich set of firm-level controls such as firm size, profitability, leverage, and sales growth in the vector $\mathbf{X}_{it}$. These variables account for differences in scale, financial capacity, and investment opportunities that may affect automation. The vector $\mathbf{W}_{ct}$ includes city-level controls, including per-capita GDP and science expenditure, which capture local development conditions and public support for innovation. Firm fixed effects, μ_i , absorb time-invariant firm characteristics, such as technology orientation, managerial style, and persistent differences in product markets. Year fixed effects, τ_t , absorb common shocks to automation demand, macroeconomic conditions,

and national policy changes.

The coefficient of interest is β . It measures whether firms located in cities with more inclusive migrant integration policies subsequently install more robots, conditional on firm characteristics, city conditions, firm fixed effects, and year fixed effects. In additional specifications, we replace the aggregate index with each policy dimension to examine which components of migrant integration policy are most closely associated with automation.

3.3 Endogeneity Concerns and Instrumental-Variable Strategy

Although the baseline specification absorbs time-invariant firm characteristics and common year shocks, the remaining city-year variation in migrant-inclusion policy may still be endogenous. Migrant-inclusion policies may be introduced alongside broader local development initiatives that independently increase firms' exposure to automation. Cities may also adjust these policies in response to wage growth or labor scarcity, creating reverse causality between local labor-market conditions and policy adoption. In addition, measurement error in our text-based policy index may bias the OLS estimate. We therefore complement the baseline analysis with an instrumental-variable strategy that predicts local migrant inclusion using policy changes in historically linguistically connected cities.

Historical dialect proximity provides a predetermined measure of longstanding cultural and informational ties across cities. Intuitively, if cities within the same dialect region expand migrant inclusion, local policymakers have greater opportunities to observe and emulate these frameworks. Migrant-inclusion policies adopted elsewhere in the dialect network should therefore predict subsequent policy choices in the focal city. Based on this logic, we construct the instrument Z_{ct} for each destination city c as the weighted sum of lagged policy indices in other cities o . The weight assigned to each source city combines its 2011 population, which scales the prominence of its policy signal, with its linguistic proximity to the focal city, which captures the strength of the historical network connection:

$$Z_{ct} = \sum_{o \neq c} \ln(\text{Population}_{o,2011}) \times \exp(-d_{oc}) \text{Inclusiveness}_{o,t-1}. \quad (4)$$

Our identification strategy relies on the exclusion restriction that dialect-network exposure affects firm robot adoption only through its effect on local migrant integration policy, conditional on the controls and fixed effects. The primary threat to this instrument is that linguistic ties also transmit migration, industrial expansion, wage changes, or technology diffusion that affect local robot demand. We therefore examine these channels from different perspectives. First, we test whether the instrument predicts local migration directly using resident-registered population gap. The relationship is statistically insignificant in the main specification, after controlling for the local

policy index, and after Frisch–Waugh–Lovell residualization. Second, we add network-weighted GDP growth, listed-firm wages, and unemployment-insurance participation growth to both stages of the IV specification. These controls capture economic activity, labor-market conditions, and formal coverage in connected source cities. The policy coefficient remains positive and statistically significant across these specifications. We discuss these results in detail in Section 4.3 and report the estimates in Appendix C.

4 Results

4.1 Baseline Results

Table 1 shows the baseline OLS estimates for firms’ robot adoption. In column (1), we include firm and year fixed effects and find a positive coefficient of 0.12 on the lagged migrant-inclusiveness index, statistically significant at the 1% level. Columns (2) and (3) sequentially add firm- and city-level controls introduced in Section 3.1. The coefficient changes only modestly to 0.11 and 0.10, respectively. The positive relationship therefore persists after accounting for differences in firm scale, financial conditions, and observable local development.

The magnitude of the Column (3) OLS estimate is economically meaningful. A one-standard-deviation increase in the policy index (5.75 units) is associated with 0.58 additional units of robot-adoption exposure in column (3). Relative to the estimation-sample mean of 22.69, this corresponds to a 2.57% increase. Given the high capital intensity and fixed costs associated with installing industrial robots, this shift represents a substantial reallocation of production technology.

The baseline estimates provide initial evidence that greater migrant inclusiveness is associated with stronger firm-level exposure to industrial robot adoption. More importantly, the result points to a distinct institutional margin of automation that is largely absent from the existing literature. Prior studies have shown that firms automate in response to explicit changes in labor prices or labor scarcity, such as minimum-wage increases, demographic shifts, and rising low-skill wages [Fan et al., 2021, Alesina et al., 2018, Dechezleprêtre et al., 2025]. Migrant-inclusive policies operate differently. Rather than directly changing a statutory wage or the quantity of available labor, they reshape the terms under which labor is employed by extending social protection, strengthening enforceable labor rights, and increasing the formal obligations associated with employment. Our findings therefore suggest that firms’ automation decisions respond not only to the market price of labor, but also to the broader institutional cost of maintaining labor relationships.

4.2 Instrumental-Variables Results

Our baseline estimation establishes the positive relationship between city-level urban inclusiveness index and automation adoption. However, the estimates may still be affected by measurement

TABLE 1 Baseline Panel Regression Results

	Industrial Robot Adoption ($Install_{ict}$)		
	(1)	(2)	(3)
Lagged Migrant-Inclusiveness Index	0.1231*** [0.0431]	0.1121*** [0.0380]	0.1016*** [0.0361]
Observations	9,258	9,160	9,117
R-squared	0.85	0.86	0.86
Outcome Mean	22.83	22.64	22.69
<i>Fixed Effects:</i>			
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
<i>Controls:</i>			
Firm Controls	–	✓	✓
City Controls	–	–	✓

Notes: This table reports OLS estimates of the relationship between the one-year-lagged migrant-inclusiveness index and firm robot-adoption exposure. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. All specifications include firm and year fixed effects. Column (1) includes no time-varying controls, column (2) adds the firm controls, and column (3) adds the city controls. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

error in the text-based policy index and by city-level shocks that jointly influence policy adoption and technology choices. Therefore, we use the dialect-network instrument described in Section 3.3 to estimate the relationship between network-predicted local migrant inclusion and robot adoption. Table 2 reports the core IV estimates.

The first stage links policies in linguistically connected cities to the focal city’s own policy choices. The result supports the spatial policy diffusion mechanism: cities with closer historical linguistic ties to regions expanding migrant rights are significantly more likely to adopt similar policies themselves (Panel B). The first-stage F -statistic is 342.50, and the Montiel Olea–Pflueger effective F -statistic is 342.45, exceeding the corresponding 10% critical value of 23.11 and rejecting weak-instrument concerns [Montiel Olea and Pflueger, 2013]. The reduced-form estimate in panel C further provides the empirical evidence by linking the same network exposure directly to more local robot exposure.

Building on this strong first stage, Panel A uses the policy variation predicted by dialect-network exposure to estimate the effect of migrant inclusion on firm automation. In Column (2), which includes the baseline firm and city controls, the 2SLS coefficient on the lagged migrant-inclusiveness index is 0.28 and is statistically significant at the 1% level. Evaluated at one standard deviation of the policy index, this estimated effect amounts to 1.60 additional units of robot adoption, equivalent to 7.07% of the sample mean. The magnitude therefore indicates a meaningful adjustment in firms’ production technology: as external policy networks push local governments to formalize migrant labor, firms respond by increasing their use of automated capital.

Finally, we assess the stability of the IV results across alternative estimators and inference procedures. Re-estimating the model using limited-information maximum likelihood (LIML), tF -adjusted inference, and Fuller(1) and Fuller(4) estimators (Appendix Table C8) yields virtually identical coefficients ranging between 0.2783 and 0.2789 [Lee et al., 2022, Fuller, 1977]. The close agreement across these alternative weak-instrument-robust procedures confirms that our positive IV relationship is stable and not driven by conventional point estimation biases.

TABLE 2 Dialect-Network IV: Main Results and Weak-Instrument Statistics

Estimator:	Industrial Robot Adoption ($Install_{ict}$)			
	(1) 2SLS No Controls	(2) 2SLS Baseline	(3) LIML	(4) tF -Adjusted 2SLS
Panel A: Second-Stage Estimates				
Lagged Migrant-Inclusiveness Index	0.4095** [0.1534]	0.2789*** [0.0947]	0.2789*** [0.0947]	0.2789*** [0.0882]
Panel B: First Stage — Dependent Variable: Lagged Migrant-Inclusiveness Index				
Dialect-Network IV (Z)	0.0034*** [0.0002]	0.0034*** [0.0002]	0.0034*** [0.0002]	— —
Panel C: Reduced Form — Dependent Variable: Industrial Robot Adoption				
Dialect-Network IV (Z)	0.0014** [0.0006]	0.0010*** [0.0003]	0.0010*** [0.0003]	— —
Panel D: Identification and Weak-Instrument Statistics				
First-Stage F	397.36	342.50	342.50	394.55
KP rk Wald F	397.36	342.45	342.45	—
MOP Effective F	397.36	342.45	342.45	—
MOP 10% Critical Value	23.11	23.11	23.11	—
First-Stage Relevance p -value	< 0.001	< 0.001	< 0.001	—
Endogeneity Test F (p)	4.35 (0.045)	3.76 (0.061)	—	—
Wu–Hausman F (p)	9.24 (0.002)	3.52 (0.061)	—	—
Conventional t Statistic	—	2.95	—	3.16
5% Critical Value	—	2.03	—	1.96
95% Confidence Interval	—	[0.086, 0.471]	—	[0.106, 0.452]
Observations	9,117	9,117	9,117	9,117
<i>Fixed Effects:</i>				
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
<i>Controls:</i>				
Firm Controls	—	✓	✓	✓
City Controls	—	✓	✓	✓

Notes: This table reports the principal IV specifications, using dialect-network exposure to instrument for the one-year-lagged migrant-inclusiveness index. Columns (1)–(2) report 2SLS estimates without and with the baseline controls, respectively; column (3) reports LIML estimates; and column (4) reports tF -adjusted inference. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The first-stage and reduced-form entries are repeated in columns (2)–(3) because the specifications use the same instrument, controls, fixed effects, and estimation sample. Column (4) reports the first-stage statistic produced by the tF procedure for the residualized model, which applies its own finite-sample variance correction. Appendix Table C8 compares the baseline 2SLS estimate with Fuller(1) and Fuller(4) estimates. Standard errors in brackets are clustered at the two-digit industry level. “—” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

4.3 Robustness and Additional Identification Evidence

Robustness of the Baseline Estimates: One concern with respect to our baseline result is that migrant integration develops alongside industrial programs that can independently encourage automation. We therefore account for the potential overlap between migrant inclusion and state-led industrial policies by adding controls for a firm’s alignment with its city’s dominant industries, the interaction of that alignment with robot-intensive manufacturing, firm participation in smart-manufacturing pilots, and city participation in the China Manufacturing 2025 program. These variables account for local industrial priorities and targeted technology support. Entered separately or jointly, they leave the policy coefficient at approximately 0.10, with the four separate-control estimates significant at 1% and the joint-control estimate at 5%. We introduce their construction in appendix B.5 and report the estimates in table C2.

To further validate our baseline results, we examine the shape of the outcome distribution, adjustment timing, and the composition of the estimation sample. PPML accommodates zero robot exposure, while two- and three-year policy lags allow for longer implementation and investment horizons. Additional geographic fixed effects absorb more local variation, and sample restrictions remove the COVID-19 years, automobile manufacturing, and the cities with the highest robot or policy exposure. The aggregate coefficient remains positive across these specifications, with magnitudes and precision varying across designs. Appendix C reports the complete estimates.

Additional Evidence on the IV Design: For the IV strategy, we first examine labor supply and economic conditions transmitted through linguistic ties. The instrument has no statistically significant relationship with the resident-registered population gap in the direct reduced form, providing no evidence of this direct migration channel. Adding network-weighted GDP growth, firm wages, and unemployment-insurance participation growth to both stages retains positive, significant policy coefficients between 0.23 and 0.28. These specifications account for observable changes in connected cities that could affect local recruitment costs and investment incentives alongside policy diffusion. Tables C7 and C10 report the estimates.

Following [Danieli et al. \[2026\]](#), we use two classes of negative controls to examine whether dialect-network exposure reflects economic conditions outside our proposed policy channel. First, the instrument is not statistically associated with the six pre-policy city characteristics, either individually after the Bonferroni adjustment or jointly. Separate pre-period regressions likewise show no systematic relationship between the instrument and earlier firm outcomes or city economic conditions. Second, future network exposure does not predict current robot adoption after conditioning on current exposure. This result holds across the linear and nonlinear specifications reported in the appendix. Together, these tests show no systematic alignment between the instrument and either economic conditions preceding the estimation period or subsequent network policy exposure.

Tables C7 and C11 report the detailed estimates.

Finally, we examine the role of network construction. Recentering source-policy shocks retains a positive and statistically significant second-stage estimate, while the observed network produces a substantially stronger first stage than the randomized-network distribution. Appendix C reports these exercises, the zero-local-policy-flow test, and the Fuller estimator comparison. Together, the results connect the IV estimates to network-predicted local policy while examining migration, connected-city economic conditions, timing, and network construction.

5 Policy Content and Firm Adjustment

5.1 Which Policy Dimensions Matter?

Migrant inclusion is a multidimensional policy package. Its components differ in how directly they affect workers' access to urban services, the obligations attached to formal employment, and the administrative costs of hiring migrant workers. To examine this variation in policy content, Table 3 replaces the aggregate index with each of its six dimensions in separate regressions.

The largest positive estimate is for public-service access. Policies expanding migrants' access to education, health care, housing, and related urban services are positively associated with robot adoption at the 1% level. Administrative burden relief displays the opposite pattern: its coefficient is negative and statistically significant at the 1% level. Social-security coordination and labor-rights protection also have positive coefficients, although both are estimated with less precision and are statistically significant at the 10% level. The estimates for hukou and residence eligibility and for employment and entrepreneurship support are not statistically distinguishable from zero. The main contrast is therefore between policies that expand access to public services and formal employment protection and policies that lower the administrative costs of employing migrant workers.

Public-service access can affect firms through two competing margins. Easier access to education, health care, and housing supports longer-term family settlement, which can reduce worker turnover and make recruitment more predictable. At the same time, a more settled workforce has stronger outside options and places greater value on wages, benefits, and stable employment terms. Access to public services is also frequently connected to employment contracts, residence documentation, or social-insurance participation. These requirements make employment relationships more formal and increase the importance of labor costs and compliance in firms' staffing decisions. The positive public-service estimate indicates that, on average, this formalization and labor-cost margin is more closely associated with firms' automation response than the potential savings from improved worker retention.

The positive estimates for social security and labor-rights protection support the same interpretation, although the statistical evidence is less precise. Social-security provisions connect migrant

employment more closely to contribution and registration requirements. Labor-rights provisions strengthen the enforcement of contracts, wage payment, equal treatment, and employment protection. Both dimensions can increase the expected cost of maintaining or adjusting a formal workforce, thereby raising the relative appeal of labor-saving capital.

Administrative burden relief operates through a different margin. By simplifying registration, documentation, approval, and other employment-related procedures, these policies reduce the fixed and matching costs of hiring migrant workers. They also make it easier for firms to recruit and adjust labor without incurring the same administrative burden. Its negative coefficient therefore indicates that administrative simplification can moderate the automation response associated with migrant inclusion. The statistically imprecise estimates for hukou and residence eligibility and for employment and entrepreneurship support are also consistent with their more indirect incidence on incumbent manufacturing firms. These policies may expand local labor supply or improve workers' economic opportunities without imposing equally direct employment obligations on firms.

This pattern complements evidence that Chinese firms respond to minimum-wage increases by substituting capital for labor [Hau et al., 2020]. Our results extend this logic from an explicit wage floor to the broader institutional content of migrant inclusion. Provisions that expand access to public services and strengthen formal employment protection are associated with greater automation, whereas measures that reduce administrative frictions are associated with less automation. These contrasts motivate the following analysis of labor expenditures, social-insurance coverage, and employment composition.

Appendix C reports the dimension-specific estimates under alternative estimators, policy lags, fixed-effect structures, and sample restrictions. Public-service access remains positively associated with robot adoption across these specifications, while administrative burden relief consistently retains a negative sign. The estimates for the remaining dimensions vary more in statistical precision.

5.2 How Do Inclusive Policies Affect Employment Costs?

The policy decomposition reveals that employment formalization is the central margin of the automation response. We therefore examine whether more inclusive policies are followed by higher labor expenditures and broader formal protection. Using firm-level financial statements from CSMAR and municipal social-protection records, Table 4 combines city unemployment-insurance participation with four firm-level outcomes: accrued compensation, cash compensation paid, wages per worker, and the firm social-insurance rate. The firm-level measures capture different components of the recurring obligations associated with employment, while the city-level measure captures the reach of formal social protection in the local labor market.

TABLE 3 Policy Decomposition

Policy Dimension:	Industrial Robot Adoption ($Install_{ict}$)					
	(1) Hukou/ Residence	(2) Social Security	(3) Public-Service Access	(4) Labor-Rights Protection	(5) Employment/ Entrepreneurship	(6) Administrative Relief
Lagged Policy-Dimension Score	0.0282 [0.1228]	0.3110* [0.1537]	0.3879*** [0.1347]	0.1266* [0.0630]	0.0585 [0.0756]	-0.4205*** [0.1535]
Observations	9,117	9,117	9,117	9,117	9,117	9,117
R-squared	0.86	0.86	0.86	0.86	0.86	0.86
Outcome Mean	22.69	22.69	22.69	22.69	22.69	22.69
<i>Fixed Effects:</i>						
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
<i>Controls:</i>						
Firm Controls	✓	✓	✓	✓	✓	✓
City Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports separate regressions of firm robot-adoption exposure on the six components of the migrant-inclusiveness index. Each policy-dimension score enters with a one-year lag. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. All specifications include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

The estimates display a consistent pattern. All four firm-level coefficients are positive: accrued compensation, wages per worker, and the firm social-insurance rate are statistically significant at the 5% level, while cash compensation is significant at the 10% level. Municipal unemployment-insurance participation also increases with the inclusive index and is significant at the 1% level. Taken together, these results show that greater migrant inclusion is accompanied by higher employment costs within firms and a broader reach of formal protection in the surrounding labor market.

These outcomes clarify the economic link between migrant inclusion and automation. Accrued compensation records employment obligations when they are recognized, whereas cash compensation captures their actual payment. Wages per worker and social-insurance contributions represent recurring costs that firms expect to bear as long as workers remain employed. An increase in these obligations therefore raises the present value of continued labor use and, consequently, the value of the labor savings generated by automated equipment. The expansion of municipal unemployment-insurance participation provides complementary evidence that this adjustment occurs as formal coverage extends more broadly across the local labor market. This interpretation is consistent with evidence that the enforcement of mandated benefits changes the cost and organization of formal employment [Almeida and Carneiro, 2012]. Prior research also documents informal

TABLE 4 Labor Costs and Social Protection

Dependent Variables:	Firm Labor Costs				City Social Protection
	(1) Accrued Compensation (mil. yuan)	(2) Cash Compensation Paid (mil. yuan)	(3) Wage per Worker (k yuan)	(4) Firm Social Insurance Rate	(5) City Unemployment Insured (10k persons)
Lagged Migrant-Inclusiveness Index	0.4948** [0.2261]	1.5153* [0.8388]	0.1629** [0.0692]	0.0001** [0.0001]	0.5829*** [0.1637]
Observations	9,102	9,117	9,117	9,058	8,871
R-squared	0.92	0.94	0.88	0.77	0.99
Outcome Mean	158.76	916.08	137.81	0.15	355.51
<i>Fixed Effects:</i>					
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
<i>Controls:</i>					
Firm Controls	✓	✓	✓	✓	✓
City Controls	✓	✓	✓	✓	✓

Notes: This table reports regressions of labor-cost and social-protection outcomes on the one-year-lagged migrant-inclusiveness index. Columns (1)–(4) use firm-level compensation, wages, and social-insurance outcomes; column (5) uses city unemployment-insurance participation matched to firm observations. The sample covers Chinese listed manufacturing firms during 2015–2024. All specifications include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

employment and payroll underreporting as adjustment margins when enforcement is incomplete [Ulyssea, 2018, Kumler et al., 2020]. By making employment relationships more documented and benefit-linked, migrant-inclusive policies can narrow these margins and ultimately incentivize firms to use labor-saving capital.

These findings also complement Wang et al. [2025], who show that robot exposure reduces corporate social-security contributions through its effects on employment, wages, and worker bargaining power. Our evidence identifies an earlier institutional margin: migrant inclusion is associated with higher compensation and broader formal protection, thereby strengthening firms' incentives to adopt labor-saving technologies.

Appendix C further examines the individual policy dimensions. Public-service access is positively associated with all three compensation measures and municipal unemployment-insurance participation. Labor-rights protection predicts higher wages per worker and broader municipal coverage, while social-security provisions are associated with cash compensation and municipal coverage. Administrative relief, by contrast, is associated with lower municipal coverage.

5.3 Which Firms Respond More Strongly?

While our estimates show that migrant inclusion accelerates automation on average, this aggregate effect naturally raises a fundamental economic question: under what specific conditions does an institutional labor shock actually translate into capital-labor substitution? The size of automation response is likely to vary across firms and institutional settings. If higher formal employment costs encourage firms to substitute capital for labor, the automation response should be larger among firms for which these costs are more consequential. The incentive to automate therefore depends on firms' reliance on labor, their ability to bear recurring employment expenses, and the regulatory environment in which they operate.

We examine this prediction along three dimensions: labor intensity, financial constraints, and the local legal environment. Labor intensity captures firms' direct exposure to increases in employment costs, while financial constraints reflect their ability to absorb these recurring expenses. Local legal environment identifies the broader institutional conditions under which labor regulations are implemented. Table 5 reports these comparisons, and Appendix B.4 describes the construction of the corresponding group indicators.

5.3.1 Labor Intensity

Firms that rely more heavily on labor are more exposed to increases in the recurring costs associated with formal employment. For a given increase in wages, benefits, or compliance obligations per worker, labor-intensive firms experience a larger increase in total production costs and gain more from replacing labor-intensive tasks with automated equipment. We therefore divide firms at the

sample median of pre-sample labor intensity and explore the heterogeneous effects on automation adoption.

Columns (1) and (2) of Table 5 show a clear difference between the two groups. Among labor-intensive firms, the coefficient on migrant inclusion is 0.15 and statistically significant at the 1% level. Among less labor-intensive firms, the estimated coefficient is close to zero. The difference between the two coefficients is statistically significant, with a p -value of 0.022. This pattern supports the employment-cost mechanism underlying the main results. When labor accounts for a larger share of production, the increase in formal employment costs induced by migrant inclusion has a greater effect on firms' overall cost structure, strengthening the incentive to adopt labor-saving technologies.

5.3.2 Financial Constraints

Financial constraints affect automation decisions through two competing forces. Firms with limited access to finance face greater difficulty funding the initial investment in robotic equipment, which can delay or discourage technology adoption. At the same time, these firms have less capacity to absorb persistent increases in wages, social-insurance contributions, and other employment-related expenses. When migrant-inclusion policies raise the expected long-term cost of labor, financially constrained firms therefore have a stronger incentive to substitute machines for labor and reduce their exposure to recurring employment costs. Their automation decisions reflect a tradeoff between the immediate financing requirements of capital investment and the longer-term savings generated by reducing labor use.

We measure financial constraints using each firm's average annual FCP score over 2010–2014 and divide firms at the cross-firm median. Higher FCP scores indicate tighter financing constraints. Our findings in Column (3) show that the automation response is concentrated among more financially constrained firms. For firms with higher FCP scores, the coefficient on migrant inclusion is 0.17, compared with only 0.03 for firms with lower scores. The difference between the two groups is statistically significant, with a p -value of 0.036.

These estimates indicate that the pressure to reduce recurring employment costs outweighs the barrier created by the initial financing needs of automation in our setting. When migrant inclusion raises expected labor costs, financially constrained firms give greater priority to technologies that reduce their future reliance on labor. The result therefore strengthens the employment-cost interpretation of our main finding: the automation response is larger among firms with less capacity to absorb sustained increases in employment expenses.

5.3.3 Legal Environment

The preceding analyses show that firms automate when they have the substitution feasibility and lack the financial advantages to absorb rising labor costs. However, the effective cost of these obligations depends on enforcement. If local legal institutions are more developed, policy provisions concerning employment contracts, social insurance, and worker protection are more likely to influence the costs firms expect to bear. Migrant-inclusive policies should therefore generate a stronger automation response in these strictly regulated locations.

We measure the local legal environment using the provincial index of market-intermediary development and legal institutions and divide firms at the annual median of the index. Consistent with our expectation, the shift toward automation is concentrated in provinces with stronger legal environments ($\beta = 0.20$), whereas firms in weaker legal regimes exhibit virtually no response ($\beta = -0.02$, with the difference significant at $p = 0.002$).

This contrast reveals how the legal environment constrains firms' outside options. In regions with weak legal institutions, firms facing higher formal employment obligations often resort to informal hiring or flexible labor arrangements to avoid compliance costs. When legal institutions are stronger, employment regulations are more consistently implemented and these alternative adjustment options become more limited. Migrant-inclusive policies are therefore more likely to translate into actual employment costs, strengthening firms' incentive to substitute capital for labor.

Appendix Table C12 reports the routine-task, product-market competition, and firms' historical violation heterogeneity. Appendix C also reports complementary labor-expenditure, overtime, and policy-dimension splits. We show that the three patterns above are robust to alternative proxies for exposure, substitution feasibility, and enforcement.

5.4 Economic Consequences of Policy-Induced Automation

Having established that migrant-inclusive policies lead firms to substitute capital for labor, we now examine the real economic consequences of this transition. Does policy-induced automation simply replace tasks, or does it alter the firm's workforce composition and risk profile? We address this by exploring two dimensions of firm-level adjustment: the skill distribution of retained workers and the structural shift in operating performance.

5.4.1 The Distributional Effect on Labor Composition

We first examine whether the automation response inadvertently reshapes the skill composition of the workforce. Migrant-inclusive policies are intended to protect vulnerable, predominantly low-skill labor. However, if these provisions raise the cost of maintaining routine employment, firms may respond by substituting away from the exact work force the policy intended to protect.

TABLE 5 Heterogeneity by Labor Intensity, Financial Constraints, and Legal Environment

Subgroups	Labor Intensity		Financial Constraints		Legal Environment	
	(1) High	(2) Low	(3) Higher FCP	(4) Lower FCP	(5) Higher	(6) Lower
Lagged Migrant-Inclusiveness Index	0.15*** [0.0403]	4.70×10^{-3} [0.0577]	0.17** [0.0646]	0.03 [0.0470]	0.20*** [0.0512]	-0.02 [0.0451]
Observations	5,226	3,891	4,545	4,572	4,256	4,779
R-squared	0.86	0.87	0.87	0.84	0.88	0.91
Outcome Mean	26.19	18.00	22.80	22.58	24.32	21.35
<i>p</i> -value for Difference	0.022**		0.036**		0.002***	
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports split-sample estimates of equation (3) for Chinese listed manufacturing firms during 2015–2024. The dependent variable is firm robot-adoption exposure, and the reported coefficient is for the one-year-lagged migrant-inclusiveness index. Columns (1)–(2) split firms by pre-sample labor intensity. Columns (3)–(4) split firms by their mean FCP score during 2010–2014; higher FCP values indicate tighter financial constraints. Columns (5)–(6) use the source data’s annual-median classification of the provincial index of market-intermediary development and legal institutions. Appendix B.4 defines the classifications and cutoffs. All regressions include the baseline firm and city controls and firm and year fixed effects. Tests of coefficient equality use 500 replications of the Stata `bdiff` procedure with the `bsample` option and seed 1234; the reported value is the procedure’s smaller-tail proportion. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6 tests this distributional implication, revealing an asymmetric employment adjustment. The policy-induced automation response is more concentrated among workers performing routine, susceptible tasks. Specifically, a one-standard-deviation increase in the migrant-inclusiveness index drives an approximately 1% decline in low-skill employment. In contrast, graduate employment remains less affected by this institutional shock.

This asymmetric adjustment highlights the unintended consequence of institutional reform. Consistent with evidence that robotics disproportionately displaces less-educated workers [Giuntella et al., 2025], our findings indicate that employment adjustment falls more heavily on low-skill workers. While inclusive policies successfully improve the formal conditions for employed migrants, they simultaneously accelerate the destruction of routine jobs as firms adopt labor-saving technologies. This pattern reveals a policy trade-off: strengthening formal employment obligations inadvertently crowds out the most vulnerable workers through capital substitution.

TABLE 6 Firm Response: Labor-Composition Reshaping

Dependent Variables:	(1) Ln(Low-Skill Workers)	(2) Ln(Graduate Workers)
Lagged Migrant-Inclusiveness Index	-0.0017** [0.0007]	0.0014 [0.0015]
Observations	7,380	8,581
R-squared	0.97	0.92
Outcome Mean	7.39	4.84
<i>Fixed Effects:</i>		
Firm FE	✓	✓
Year FE	✓	✓
<i>Controls:</i>		
Firm Controls	✓	✓
City Controls	✓	✓

Notes: This table reports regressions of firm labor composition on the one-year-lagged migrant-inclusiveness index. The dependent variables are the logarithms of low-skill and graduate employment. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. Both specifications include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

5.4.2 Operating Leverage and Firm Performance

Beyond workforce restructuring, substituting labor with industrial robots simultaneously changes a firm's cost structure. In traditional manufacturing, labor functions largely as a variable cost. Industrial robots, however, represent substantial fixed capital commitments. By automating, firms replace part of their ongoing labor expenses with fixed capital costs, thereby increasing their operating leverage.

Table 7 confirms this structural tradeoff by interacting the migrant-inclusiveness index with

robot exposure. In firms with greater robot exposure, increases in migrant inclusion are associated with fewer employees, higher production efficiency, and more volatile profits. Evaluated at a one-standard-deviation increase in both policy and robot exposure, firms reduce an average of 93 employees and realize a 2.15% gain in overall production efficiency. However, this leaner operation comes at the cost of stability, generating a 2.34% increase in profit volatility.

This combination of higher production efficiency and greater profit volatility is consistent with increased operating leverage. While automation streamlines production and shields firms from recurring labor compliance costs, a higher fixed-cost base makes corporate earnings significantly more sensitive to changes in sales and capacity utilization. Complementing existing literature that views robot adoption purely as a productivity-enhancing mechanism [Koch et al., 2021, Acemoglu et al., 2020], our findings illustrate the financial boundaries of this transition. When institutional pressures force firms to automate, they emerge leaner and more efficient, but inherently more exposed to business cycle fluctuations.

TABLE 7 Firm Response: Employment, Production Efficiency, and Profit Volatility

Dependent Variables:	(1) Employee Count	(2) DEA Efficiency	(3) Profit Volatility (bp)
Lagged Migrant-Inclusiveness Index $\times$ Industrial Robot Adoption	-0.46** [0.1837]	2.89×10^{-5} *** [4.70×10^{-6}]	0.03* [0.0182]
Observations	9,117	9,115	9,117
R-squared	0.97	0.76	0.48
Outcome Mean	6424.15	0.27	289.33
Outcome Std. Dev.	8901.65	0.22	335.71
One-SD Interaction Contrast	-92.59	0.0058	6.76
Relative Effect (%)	-1.44	2.15	2.34
<i>Fixed Effects:</i>			
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
<i>Controls:</i>			
Baseline Controls	✓	✓	✓

Notes: This table reports regressions of firm employment, production efficiency, and profit volatility on the migrant-inclusiveness index, robot exposure, and their interaction. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. DEA efficiency is measured on a zero-to-one scale, and profit volatility is measured in basis points. All outcomes are winsorized at the 2nd and 98th percentiles within their available baseline samples. Each specification includes the baseline controls and firm and year fixed effects. The one-SD interaction contrast, $\hat{\beta}_{XI\sigma_X\sigma_I}$, measures the difference in the estimated policy response across a one-standard-deviation difference in robot exposure. The relative effect divides this contrast by the corresponding outcome mean. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

6 Conclusion

What happens when cities give migrant workers access to public services, social insurance, and enforceable labor rights? The conventional answer is stabilization: migrants stay longer, firms recruit more easily, and labor shortages ease. This paper shows the opposite. On the firm side, migrant inclusion leads manufacturers to adopt more robots and rely less on low-skill labor. By formalizing employment relationships that were previously more flexible, policies intended to support migrant workers also change firms' incentives to substitute capital for labor.

We construct a city-year migrant-inclusiveness index from 3,408 policy records assigned to cities during 2014–2023 and combine it with a panel of Chinese listed manufacturing firms observed from 2015 to 2024. Using a dialect-network instrumental variable strategy, we find that a one-standard-deviation increase in migrant inclusion raises robot exposure by approximately 1.60 units, or 7.07% of the sample mean. Our analyses show that this automation response is driven by a labor-cost and compliance channel. Expanded rights and public services make employment relationships more formal, more rigid, and more visible to regulators. Firms therefore respond by adopting more robots, substituting heavily regulated human labor with predictable capital investments. The policy decomposition reinforces this interpretation: public-service access and labor-rights protection are associated with greater robot adoption, whereas administrative burden relief reduces the automation response. The technological consequences of inclusion therefore depend not only on the rights being extended, but also on how those rights are implemented.

Consistent with this compliance-driven mechanism, the shift toward capital is highly targeted. The adjustment is strongest among firms for which the change in employment conditions matters most. Labor-intensive firms have more workers whose tasks can be replaced, financially constrained firms have less capacity to absorb continuing employment expenses, and firms in provinces with stronger legal institutions face more effective enforcement of formal obligations. These firms exhibit the largest automation responses. The resulting reorganization extends beyond technology adoption. Low-skill employment declines, while the interaction between migrant inclusion and robot exposure is associated with higher DEA production efficiency and greater profit volatility. Together, these findings describe a shift toward a more capital-intensive mode of production: firms operate with less low-skill labor and greater measured efficiency, but they also face greater variation in profits as fixed capital becomes more important.

These results reveal a central paradox of labor-market inclusion: policies that make cities more inclusive for migrant workers can make factory employment less inclusive. Yet the negative relationship between administrative burden relief and automation also implies that this trade-off is not inevitable. Policy design can reduce it by pairing expanded migrant rights with simpler administrative procedures, more portable social-insurance arrangements, and support for workers moving into tasks that complement new technologies. Where possible, access to public services

and social protection should be separated from avoidable firm-level procedures that make each employment relationship harder to maintain. The workplace consequences of migrant inclusion therefore depend on how worker protections are combined with measures that facilitate firms' adjustment.

Appendices

Appendix A: Construction of City-Level Migrant-Inclusive Policy Index

A.1 Policy Sources and Sample Selection

We construct six dimension-specific measures to capture the institutional evolution of migrant-inclusive labor policies at the city level. The policy texts are collected from the PKULaw local-policy database. The search covers provincial, municipal, and district or county policies related to the floating population, household registration (*hukou*), the non-*hukou* population, non-local labor, and migrant workers.

The policy collection spans 2000–2024. The estimation window uses 3,408 records assigned to cities during 2014–2023, representing 3,408 distinct source policy identifiers. Each record identifies its issuer and geographic scope. Forwarding documents issued by different localities constitute separate records. The index combines historical policy scores with AI-assisted coding and manual review of additional documents. Each record receives one overall score and six separately assigned dimension scores; the overall score is not the sum of the six dimensions. Scores are based on substantive provisions when the full text is available and on policy titles otherwise.

A.2 Coding Dimensions and Scoring Rules

The index measures policy content along six distinct institutional domains:

1. **Hukou and residence eligibility:** Settlement, residence permits, and identity-based treatment.
2. **Social security and welfare coordination:** Pension, medical, unemployment, maternity, work-injury insurance, and portability.
3. **Public-service access:** Education for migrant children, public health, subsidized housing, and legal aid.
4. **Labor-rights protection:** Wage payment, minimum wage, labor contracts, equal pay, and labor inspection.
5. **Employment and entrepreneurship support:** Vocational training, start-up subsidies, and employment services.
6. **Administrative burden relief and control:** Residence registration, inspection, penalties, fees, and forced removal.

Each dimension is scored on a five-point scale from -2 to 2 based on the institutional scope and duration of the provisions. Table A1 summarizes the common scoring criteria across all dimensions. The dimensions are not mutually exclusive; a single policy record may receive nonzero scores in several dimensions.

Table A1: Common Document-Level Scoring Scale

Score	Institutional access, protection, and support (Dim. 1–5)	Administrative burden relief (Dim. 6)
2	Broad and institutionalized inclusive arrangements.	Repeal of discriminatory controls; institutionalized equal treatment.
1	Temporary, targeted, pilot, or one-time inclusive measures.	Procedural simplification; reduced documentation or fees.
0	Neutral, unrelated, or purely technical provisions.	Routine statistics; uniform technical administration.
-1	Temporary restrictions or short-term barriers to access.	Temporary migrant-specific checks, extra documentation, or penalties.
-2	Persistent, institutionalized, or discriminatory exclusion or restriction.	Persistent migrant-specific control, punitive obligations, or monitoring.

Note: For the administrative burden dimension, positive scores indicate relief or reduced burdens, whereas negative scores indicate stronger migrant-specific controls.

A.3 City-Year Aggregation and Empirical Treatment

After assigning document-level scores, we sum the scores within each city-year and dimension to obtain six city-year subindices. Let $\mathcal{D}_{ct}$ denote the set of policy records assigned to city c in policy year t , and let $s_{d,k}$ denote the score of policy d in dimension k . The city-year index is defined as

$$Index_{c,t,k} = \sum_{d \in \mathcal{D}_{ct}} s_{d,k}.$$

City-years without relevant policy records receive zero. Provincial records contribute to cities within their recorded geographic scope, municipal records to the issuing city, and district or county records to the parent city. We sum the overall scores to form the aggregate index and sum the scores for each dimension separately to form the corresponding subindex. The resulting indices measure annual policy flows, using each record’s effective or assigned policy year.

The one-year lag maps policy scores from 2014–2023 to firm outcomes from 2015–2024. Policy scores and robot exposure are winsorized at the 2nd and 98th percentiles of their full firm-panel distributions. For the two- and three-year lag specifications, we shift the winsorized policy series by two and three years, respectively, while retaining only policy years within the same 2014–2023 window.

Appendix B: Supplementary Data and Instrument Construction

B.1 Data Sources, Matching, and Estimation Samples

The analysis combines the city-level policy panel with Chinese listed manufacturing firms observed during 2015–2024. City policies enter with a one-year lag, so the relevant policy-year window is 2014–2023. Firm financial statements and financial controls are drawn from CSMAR, and the firm records also incorporate Wind information on employment structure and governance. Municipal statistical sources provide city economic conditions and social-protection outcomes. Firm records are matched by stock code and calendar year; city information is attached by city and year.

The robot-exposure measure combines three inputs. Annual Chinese industry robot installations come from the International Federation of Robotics. Industry employment in 2010 comes from the *China Industry Statistical Yearbook*. Firm production-worker shares in 2011 come from Wind. Equation (2) first divides industry installations by baseline industry employment and then applies the firm’s production-worker share normalized by the cross-firm median. The employment denominator and firm exposure weight are fixed before the main outcome period, while industry robot installations vary annually.

Table C1 reports variable-specific complete samples. The robot outcome and aggregate policy index each have 9,258 observations in the descriptive sample, while the baseline regression with the complete control set contains 9,117 firm-year observations. Other regressions use the available observations for their outcome or grouping variable. For example, the labor-cost and employment-composition regressions have different sample sizes because their respective accounting and employee records differ in coverage. Each table reports the estimation sample actually used.

Policy lags are formed at the city level before the policy data are matched to firm observations. The aggregate policy index, its six dimensions, and robot exposure are winsorized at the 2nd and 98th percentiles of their corresponding full firm-panel distributions. Additional firm-outcome variables are winsorized within their available baseline samples. Variables used for subgroup assignment retain their original grouping values.

B.2 Firm and City Covariates

The firm control vector contains size, leverage, return on assets, asset turnover, sales growth, firm age, and log employment. These variables account for differences in scale, balance-sheet conditions, profitability, and investment opportunities. Governance and ownership controls include state ownership, board size, board independence, CEO-chair duality, the largest shareholder’s ownership share, and Big Four auditor status. The city control vector includes per-capita GDP and science expenditure. These covariates enter the baseline specification alongside firm and year fixed effects, as indicated in the individual tables.

The reported monetary units are retained throughout the analysis. City per-capita GDP is measured in yuan per person, and city science expenditure in units of 10,000 yuan. Firm employment enters the control vector in logarithms, while the employee-count outcome is reported in persons. Maintaining this distinction is necessary when interpreting coefficients across the employment tables.

B.3 Labor Costs, Social Protection, and Firm Outcomes

The cost analysis uses accrued compensation and cash compensation paid, both reported in million yuan; wages per worker, reported in thousand yuan; the firm social-insurance rate; and city unemployment-insurance participation, reported in units of 10,000 persons. Accrued and cash-paid compensation distinguish recognized labor obligations from disbursements. Wages per worker describe remuneration at the worker level. The social-insurance rate and municipal participation count provide firm and city measures of formal social protection. These series enter separate regressions under the baseline control and fixed-effect structure.

The labor-composition analysis uses log low-skill employment and log graduate employment. The firm-outcome analysis uses employee count, DEA production efficiency, and profit volatility in basis points. DEA efficiency is the total-efficiency score, calculated within industry-year groups using an input-oriented, constant-returns-to-scale model. Revenue is the output; inputs include fixed assets, capitalized R&D, goodwill, intangible assets, inventory, and selling plus administrative expenses. Higher values indicate greater efficiency.

The firm-outcome regressions include the policy index, robot exposure, and their interaction. Table 7 reports the interaction contrast $\hat{\beta}_{XI}\sigma_X\sigma_I$, which compares the estimated response to a one-standard-deviation policy change across a one-standard-deviation difference in robot exposure. DEA efficiency is observed for 9,115 baseline firm-year observations.

B.4 Heterogeneity Classifications

The main heterogeneity analysis uses three classifications. Labor intensity is split at its pre-sample median. Financial constraints are measured by the FCP index: annual firm scores are averaged over 2010–2014 and split at the median across 940 firms (0.490305). Higher FCP values indicate tighter financial constraints.

The local legal environment is measured by the provincial index of market-intermediary development and legal institutions, matched to firms by year. We retain the high/low classification supplied with the source data, which is based on the annual median and preserves the source treatment of observations tied at that median. Unlike the pre-sample firm classifications, the legal-environment classification varies over time. The split-sample regressions contain 4,256 observations in the higher-index group and 4,779 in the lower-index group.

The supplementary heterogeneity analysis includes prior corporate violations, the pre-sample routine low-skill share, and two measures of product-market competition. The prior-violation indicator equals one if an announcement dated between 2010 and 2014 records a violation by the listed company itself (`IsViolated=Y`), and zero otherwise. The underlying records come from announcements issued by securities regulators, stock exchanges, other government agencies, and listed companies. We deduplicate firm–event records and keep this pre-sample classification fixed during 2015–2024. The data identify 272 firms with prior violation records and 670 without such records, corresponding to 2,614 and 6,503 regression observations.

The Boone-type competition measure is the industry-year slope from a regression of log market share on log average cost, where average cost is operating cost divided by operating revenue. It is estimated within CSRC 2012 two-digit industries with at least 15 firms per industry-year. The signed slope defines the index, and the split is made at the pooled firm-year median of 1.566949. The firm-markup split uses the `Markup_DLWTL` measure and a pooled firm-year median of 1.066467; lower markups indicate stronger competitive pressure. Additional classifications cover labor expenditures, overtime, political connections, distance to government, shareholder-meeting frequency, and environmental-policy pressure.

All split-sample regressions include the baseline controls and firm and year fixed effects. The Stata `bdiff` procedure compares group coefficients using 500 replications, the `bsample` option, and seed 1234. The tables report the empirical smaller-tail proportion, $\min(f/500, 1 - f/500)$, where f is the resampling frequency returned by the command. Differences in group sample sizes reflect the observations available for each classification.

B.5 Industrial-Policy Controls

Four controls account for local industrial priorities and targeted support for technological upgrading. The first, *dominant-industry match*, equals one when a firm’s current CSRC 2012 two-digit industry appears in its city’s dominant-industry list from the preceding year. The source is a city-year panel derived from the *City 2000–2024 Historical Dominant Industries* file. The city lists are cumulative, and the analysis uses the lagged entries required for the 2014–2023 policy window.

The second control interacts this lagged match with a fixed robot-intensive-industry indicator. That indicator equals one for CSRC 2012 codes C13–C16, C25–C29, C31–C33, C36, and C38–C40, and zero otherwise. The interaction identifies firms that both match the local dominant-industry list and operate in a robot-intensive industry under this mapping.

The third control is a firm-year smart-manufacturing pilot indicator, constructed as the treatment indicator interacted with the post-selection indicator and matched by firm and year. The fourth is a city-year China Manufacturing 2025 pilot indicator that switches on after the city enters the program and is matched by prefecture-city code and year. Table C2 adds each control separately

and then all four jointly.

B.6 Dialect-Network Weights and First Stage

The instrument combines historical linguistic proximity with policy issuance in other cities. Equation (4) defines the fixed weight using the source city’s 2011 population and its dialect distance from the destination city. The weight declines with distance and increases with baseline population. Source cities enter with their lagged migrant-inclusiveness indices. The focal city is excluded, and the weighted sum in equation (4) is unnormalized.

The historical distance and 2011 population components hold the network weights fixed over the outcome period. Time variation comes from the policy indices of connected source cities. For a given change in source-city policy, the contribution to the focal city’s instrument is greater when the cities are linguistically closer or the source city has a larger baseline population.

The first-stage regression relates local inclusiveness to this exposure using the baseline controls and fixed effects. Table 2 reports the first stage, second stage, reduced form, LIML estimates, and tF inference. Table C8 adds the Fuller estimator comparison.

B.7 Additional Identification Checks

The supplementary tests follow the economic channels through which linguistic connections could affect automation alongside local policy. For the labor-supply channel, the migration proxy is the resident–registered population gap share, a stock-based measure of local population mobility. Table C7 reports its direct reduced-form relationship with the instrument, a specification conditioning on the policy index, and a Frisch–Waugh–Lovell residualized specification. These regressions examine whether network exposure is associated with the local population composition relevant to recruitment and labor availability.

For the transmission of economic conditions, Table C10 adds network-weighted GDP growth, listed-firm wages, and unemployment-insurance participation growth to both stages. These variables capture activity, remuneration, and formal coverage in connected source cities using the network exposure structure. A separate zero-local-policy-flow exercise estimates reduced-form relationships within observations whose lagged local policy index is zero, examining whether the instrument remains associated with robot exposure, wages, or the population gap when measured local policy flow is absent.

The timing checks examine the alignment of the instrument with earlier economic conditions and subsequent policy exposure. Table C7 reports pre-policy city-characteristic negative controls, with a Bonferroni adjustment for the individual tests and a joint test. It also tests future network exposure conditional on realized exposure using linear and quadratic specifications and a quadratic specification augmented with a current-IV–future-IV interaction. Table C11 relates the 2015 network

instrument to firm robot-exposure trends during 2012–2014 and firm characteristics in 2014, as well as city trajectories in per-capita GDP, science expenditure, and GDP growth during 2012–2014. Firm tests include province and industry fixed effects; city tests include province fixed effects.

Finally, the network-construction checks recenter source policy shocks and compare the observed first stage with the distribution generated by randomly reassigning source-city policy identities across the network. This comparison examines how the association with local policy changes when the observed pairing of source policies and network positions is disrupted. Appendix C collects the estimates, with samples and standard-error conventions reported in the table notes.

Appendix C: Supplementary Tables and Figures

C.1 Summary Statistics

Table C1 reports summary statistics for the policy measures, robot exposure, baseline controls, mechanism outcomes, and heterogeneity variables used in the empirical analysis.

TABLE C1 Descriptive Statistics for the Estimation Variables

Variable	Obs.	Mean	Std. Dev.	Min	Median	Max
Panel A: Outcome and Policy						
Industrial Robot Adoption ($Install_{ict}$)	9,258	22.83	35.24	0.00	8.96	157.78
Lagged Migrant-Inclusiveness Index ($Inclusiveness_{c,t-1}$)	9,258	6.70	5.73	-1.00	6.00	23.00
Panel B: Policy Dimensions						
Hukou/Residence Score	9,117	1.01	1.40	0.00	0.00	5.00
Social Security Score	9,117	1.32	1.66	0.00	1.00	6.00
Public Service Score	9,117	2.54	2.54	0.00	2.00	10.00
Labor Rights Score	9,117	6.40	5.12	0.00	6.00	21.00
Employment/Entrepreneurship Score	9,117	1.61	2.07	0.00	1.00	8.00
Administrative-Relief Score	9,117	-1.67	2.60	-10.00	-1.00	2.00
Panel C: Firm Characteristics						
Size	9,160	22.66	1.22	19.74	22.50	26.45
Leverage	9,160	0.42	0.18	0.05	0.42	0.93
ROA	9,160	0.04	0.06	-0.37	0.03	0.26
Sales Growth	9,160	0.12	0.31	-0.65	0.08	3.81
Ln(Employment)	9,160	8.13	1.14	4.45	8.05	11.18
SOE	9,160	0.37	0.48	0.00	0.00	1.00
Panel D: City Characteristics						
City Per Capita GDP (yuan/person)	9,117	113922.01	48046.31	16121.00	110067.00	285338.00
City Science Expenditure (10k yuan)	9,117	1197372.60	1611914.50	753.00	355868.00	6354851.00
Panel E: Labor Costs, Social Protection, and Employment						
Accrued Compensation (mil. yuan)	9,102	158.76	318.22	2.05	51.91	1839.95
Wage per Worker (k yuan)	9,117	137.81	60.64	57.59	124.23	336.86
Firm Social Insurance Rate	9,063	0.15	0.05	0.05	0.15	0.29
City Unemployment Insured (10k persons)	8,871	355.51	407.78	13.69	146.00	1359.01
Ln(Low-Skill Workers)	7,405	7.39	1.21	4.73	7.34	10.09
Ln(Graduate Workers)	8,593	4.84	1.62	1.61	4.83	8.31
Panel F: Grouping Variables for Heterogeneity						
Pre-Sample Labor-Intensive (=1)	9,117	0.57	0.49	0.00	1.00	1.00
FCP Score, 2010–2014 Mean	9,117	0.47	0.19	0.02	0.49	0.84
Pre-Sample Violation Record (=1)	9,117	0.29	0.45	0.00	0.00	1.00
Routine Low-Skill Share	9,088	0.68	0.15	0.03	0.70	0.93
Annual Overtime Intensity	9,108	0.34	0.14	0.00	0.34	0.81
Distance to Government (km)	9,117	21.30	20.32	0.39	14.75	251.68

Notes: This table reports descriptive statistics for the outcome, policy measures, covariates, labor outcomes, and heterogeneity variables used in the analysis. The unit of observation is the firm-year in the 2015–2024 panel of Chinese listed manufacturing firms; city-level variables are matched to firm observations. Panel A reports robot-adoption exposure and the aggregate policy index. Robot exposure combines industry-level robot installations with pre-sample employment weights, as defined in equation (2). The policy measures are constructed from documents issued during 2014–2023 and enter the regressions with a one-year lag. Panel B reports the six policy subindices; positive values indicate greater inclusion, whereas negative administrative-relief scores indicate additional administrative burdens. Panels C–D report firm and city covariates. Panel E reports compensation, insurance, and employment outcomes in the units shown. Panel F reports the heterogeneity variables. The FCP score is averaged within firms over 2010–2014, and the violation indicator identifies companies with a violation announced during the same period. Observation counts reflect the available sample for each variable and may differ from the regression samples. Continuous outcomes and policy indices are winsorized at the 2nd and 98th percentiles; the grouping variables retain their original assignment values. Appendix B provides additional variable definitions.

C.2 Baseline Robustness and Sensitivity

The following tables examine concurrent industrial policies, alternative estimators and policy lags, additional fixed effects, sample restrictions, and alternative winsorization thresholds. Figure C1 complements these specifications by reporting the sensitivity of the baseline estimate to an omitted confounder benchmarked to city per-capita GDP.

TABLE C2 Migrant Inclusion and Robot Exposure: Controlling for Industrial Policies

Added Control:	Industrial Robot Adoption ($Install_{ict}$)				
	(1) Dominant Industry Match	(2) Dominant $\times$ Robot- Intensive	(3) Smart Mfg. Pilot	(4) CM2025 Pilot City	(5) All Four Controls
Lagged Migrant-Inclusiveness Index	0.1022*** [0.0366]	0.1012*** [0.0368]	0.1028*** [0.0363]	0.0987*** [0.0361]	0.0982** [0.0363]
Observations	9,117	9,117	9,117	9,117	9,117
R-squared	0.8564	0.8566	0.8565	0.8565	0.8567
Outcome Mean	22.6927	22.6927	22.6927	22.6927	22.6927
<i>Fixed Effects:</i>					
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
<i>Controls:</i>					
Firm Controls	✓	✓	✓	✓	✓
City Controls	✓	✓	✓	✓	✓
Dominant-Industry Match	✓	–	–	–	✓
Dominant $\times$ Robot- Intensive Industry	–	✓	–	–	✓
Smart-Manufacturing Pilot	–	–	✓	–	✓
CM2025 Pilot City	–	–	–	✓	✓

Notes: This table reports estimates of equation (3) with additional controls for industrial policies that may reflect local production priorities or targeted support for automation. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure, and the reported policy coefficient is for the one-year-lagged migrant-inclusiveness index. Column (1) adds an indicator for whether the firm’s two-digit industry appears in its city’s cumulative dominant-industry list in the preceding year. Column (2) adds the interaction of this indicator with robot-intensive manufacturing, defined by CSRC 2012 codes C13–C16, C25–C29, C31–C33, C36, and C38–C40. Column (3) adds the firm-level smart-manufacturing-pilot treatment-by-post indicator. Column (4) adds an indicator that turns on after the city enters the China Manufacturing 2025 pilot program. Column (5) includes all four controls. Appendix B.5 describes the data sources and matching procedures. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C3 Migrant Inclusion and Robot Exposure: Estimator, Timing, and Sample Checks

	Industrial Robot Adoption ($Install_{ict}$)					
Specification:	(1) Baseline OLS	(2) PPMLHDFE	(3) OLS, Lag $t - 2$	(4) OLS, Lag $t - 3$		
Panel A: Estimator and Timing						
Lagged Migrant-Inclusiveness Index	0.1016*** [0.0361]	0.0033*** [0.0006]	0.0967*** [0.0350]	0.0772* [0.0418]		
Observations	9,117	8,112	8,231	7,330		
R-squared	0.86	–	0.88	0.91		
Outcome Mean	22.69	25.50	24.16	25.81		
Firm FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		
Firm Controls	✓	✓	✓	✓		
City Controls	✓	✓	✓	✓		
	Industrial Robot Adoption ($Install_{ict}$)					
Specification:	(1) Province– Year FE	(2) City FE	(3) COVID Excluded	(4) C36 Auto Excluded	(5) Drop Top-5 Robot Cities	(6) Drop Top-5 Policy Cities
Panel B: Fixed Effects and Sample Composition						
Lagged Migrant-Inclusiveness Index	0.2131* [0.1112]	0.0872** [0.0352]	0.1809*** [0.0615]	0.1026*** [0.0348]	0.1061*** [0.0358]	0.1334*** [0.0451]
Observations	9,116	9,116	6,305	8,623	8,995	8,521
R-squared	0.87	0.86	0.83	0.77	0.85	0.86
Outcome Mean	22.69	22.68	19.59	18.67	21.94	21.98
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	–	✓	✓	✓	✓	✓
Province–Year FE	✓	–	–	–	–	–
City FE	–	✓	–	–	–	–
Baseline Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports alternative specifications of equation (3) that vary the outcome model, policy timing, fixed effects, and estimation sample. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure. In Panel A, column (1) reports baseline OLS estimates with a one-year policy lag, column (2) uses Poisson pseudo-maximum likelihood with high-dimensional fixed effects (PPMLHDFE), and columns (3)–(4) use two- and three-year policy lags. The PPML coefficient is a semi-elasticity; the remaining coefficients are measured in robot-exposure units. In Panel B, column (1) replaces year fixed effects with province-by-year fixed effects, and column (2) adds city fixed effects. Columns (3)–(6) exclude the COVID-19 years, automobile manufacturing (C36), the five cities with the highest robot exposure, and the five cities with the highest policy index, respectively. All specifications include the baseline controls; the included fixed effects are indicated in the table. Observation counts reflect the estimation sample for each specification. Standard errors in brackets are clustered at the two-digit industry level. “—” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C4 Policy-Dimension Estimates under Alternative Estimators and Lags

Policy Dimension:	Industrial Robot Adoption ($Install_{ict}$)					
	(1) Hukou/ Residence	(2) Social Security	(3) Public-Service Access	(4) Labor-Rights Protection	(5) Employment/ Entrepreneurship	(6) Administrative Relief
Panel A: Baseline OLS, Lag $t - 1$						
Lagged Policy-Dimension Score	0.0282	0.3110*	0.3879***	0.1266*	0.0585	-0.4205***
	[0.1228]	[0.1537]	[0.1347]	[0.0630]	[0.0756]	[0.1535]
Observations	9,117	9,117	9,117	9,117	9,117	9,117
R-squared	0.86	0.86	0.86	0.86	0.86	0.86
Outcome Mean	22.69	22.69	22.69	22.69	22.69	22.69
Panel B: PPMLHDFE, Lag $t - 1$						
Lagged Policy-Dimension Score	0.0046	0.0161***	0.0129***	0.0050***	0.0070***	-0.0092***
	[0.0046]	[0.0048]	[0.0037]	[0.0017]	[0.0017]	[0.0025]
Observations	8,112	8,112	8,112	8,112	8,112	8,112
R-squared	—	—	—	—	—	—
Outcome Mean	25.50	25.50	25.50	25.50	25.50	25.50
Panel C: OLS, Lag $t - 2$						
Lagged Policy-Dimension Score	0.1557	0.2614**	0.2756**	0.1486*	-0.0495	-0.3142***
	[0.1139]	[0.1067]	[0.1030]	[0.0747]	[0.0572]	[0.1036]
Observations	8,231	8,231	8,231	8,231	8,231	8,231
R-squared	0.88	0.88	0.88	0.88	0.88	0.88
Outcome Mean	24.16	24.16	24.16	24.16	24.16	24.16
Panel D: OLS, Lag $t - 3$						
Lagged Policy-Dimension Score	0.1148	0.2249**	0.1814**	0.1265**	0.0499	-0.2273***
	[0.0899]	[0.1023]	[0.0766]	[0.0577]	[0.0489]	[0.0627]
Observations	7,330	7,330	7,330	7,330	7,330	7,330
R-squared	0.91	0.91	0.91	0.91	0.91	0.91
Outcome Mean	25.81	25.81	25.81	25.81	25.81	25.81
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Firm Controls	✓	✓	✓	✓	✓	✓
City Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports separate regressions in which each of the six policy subindices replaces the aggregate index in equation (3). The specifications vary the estimator and policy lag. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure. Columns (1)–(6) correspond to hukou and residence eligibility, social security, public-service access, labor-rights protection, employment and entrepreneurship support, and administrative burden relief. Panel A uses OLS with a one-year policy lag. Panel B uses PPMLHDFE with the same lag; its coefficients are semi-elasticities. Panels C–D use OLS with two- and three-year policy lags, respectively. Appendix A defines the policy subindices. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. “—” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C5 Policy-Dimension Estimates under Alternative Fixed Effects and Samples

Specification:	(1) Baseline FE	(2) Province–Year FE	(3) City FE		
Panel A: Alternative Fixed-Effect Structures					
Hukou/Residence	0.0282 [0.1228]	0.3078 [0.2921]	0.0122 [0.1532]		
Social Security	0.3110* [0.1537]	0.5409 [0.4304]	0.2939** [0.1367]		
Public-Service Access	0.3879*** [0.1347]	1.0761*** [0.3033]	0.3774*** [0.1284]		
Labor-Rights Protection	0.1266* [0.0630]	0.4512** [0.2033]	0.1111** [0.0540]		
Employment/Entrepreneurship	0.0585 [0.0756]	0.4039 [0.3130]	0.0375 [0.0713]		
Administrative Relief	-0.4205*** [0.1535]	-0.6616** [0.2867]	-0.4140*** [0.1506]		
Observations	9,117	9,116	9,116		
R-squared	0.86	0.87	0.86		
Outcome Mean	22.69	22.69	22.68		
Firm Controls	✓	✓	✓		
City Controls	✓	✓	✓		
Firm FE	✓	✓	✓		
Year FE	✓	–	✓		
Province–Year FE	–	✓	–		
City FE	–	–	✓		
	(1)	(2)	(3)	(4)	(5)
Specification:	Baseline	No COVID	No C36	Drop Top-5 Install Cities	Drop Top-5 Inclusive Cities
Panel B: Sample Restrictions					
Hukou/Residence	0.0282 [0.1228]	-0.0049 [0.1499]	0.0047 [0.1267]	0.0375 [0.1259]	0.0041 [0.1153]
Social Security	0.3110* [0.1537]	0.3830* [0.2040]	0.3014* [0.1496]	0.3300** [0.1586]	0.3721* [0.1894]
Public-Service Access	0.3879*** [0.1347]	0.4822** [0.2081]	0.3756*** [0.1271]	0.4066*** [0.1358]	0.4499** [0.1867]
Labor-Rights Protection	0.1266* [0.0630]	0.1723 [0.1088]	0.1302** [0.0604]	0.1294** [0.0594]	0.0954 [0.0664]
Employment/Entrepreneurship	0.0585 [0.0756]	0.0850 [0.1293]	0.0910 [0.0685]	0.0662 [0.0694]	0.1008 [0.0957]
Administrative Relief	-0.4205*** [0.1535]	-0.4951** [0.2163]	-0.4235*** [0.1439]	-0.4281** [0.1576]	-0.3371* [0.1700]
Observations	9,117	6,305	8,623	8,995	8,521
R-squared	0.86	0.83	0.77	0.85	0.86
Outcome Mean	22.69	19.59	18.67	21.94	21.98
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓	✓

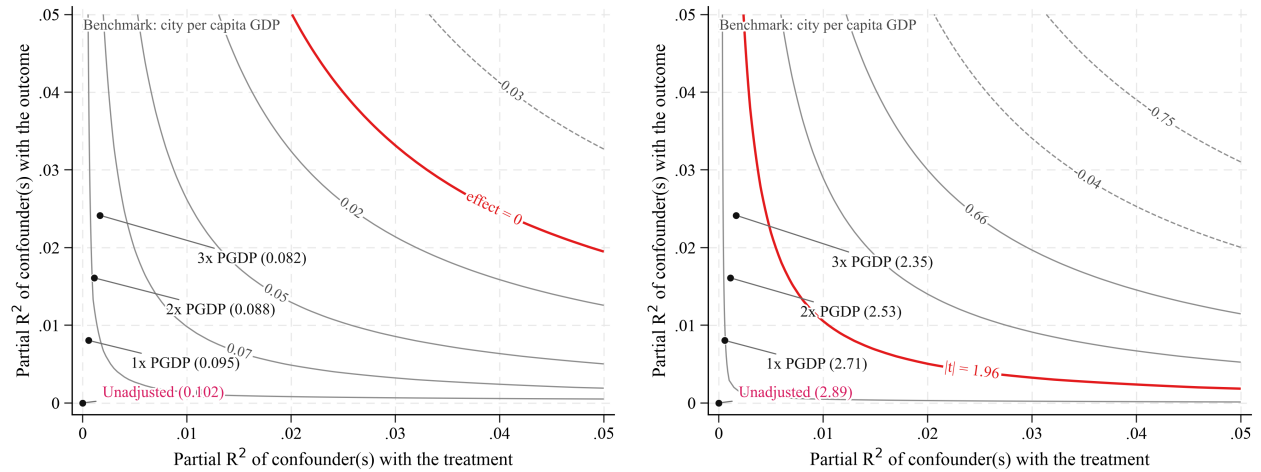
Notes: This table reports policy-dimension estimates under alternative geographic fixed effects and sample restrictions. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure. Each coefficient comes from a separate regression based on equation (3), using the indicated policy subindex with a one-year lag. In Panel A, column (1) uses the baseline fixed effects, column (2) replaces year fixed effects with province-by-year fixed effects, and column (3) adds city fixed effects. Panel B reports the baseline sample in column (1); columns (2)–(5) exclude the COVID-19 years, automobile manufacturing (C36), the five cities with the highest robot exposure, and the five cities with the highest policy index, respectively. All regressions include the baseline firm and city covariates; the included fixed effects are indicated in the table. Each column contains six separate regressions. The reported R-squared is their common value rounded to two decimals, or a range when the rounded values differ. Standard errors in brackets are clustered at the two-digit industry level. “–” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C6 Industry Composition of Upper-Tail Robot Exposure

Industry Code	Industry Name	Ind	Clipped Obs.	Share of Clipped Sample
C36	Automobile manufacturing	94	176	95.1%
C37	Railway, ship, aerospace, and transport equipment	95	4	2.2%
C39	Computer, communication, and electronic equipment	100	4	2.2%
C38	Electrical machinery and equipment	99	1	0.5%

Notes: This table reports the industry distribution of firm-year observations above the 98th-percentile cutoff of robot-adoption exposure before winsorization in the 2015–2024 listed-manufacturing sample. Industry codes follow the CSRC 2012 classification; Ind is the corresponding numerical identifier. Clipped Obs. is the number of observations affected by the upper-tail cap, and Share is the industry’s percentage of all affected observations. Percentages are rounded. Table C3 separately reports estimates that exclude automobile manufacturing. Winsorization caps values without removing observations; this table contains no regression estimates.

Figure C1: Sensitivity of the Baseline Estimate to an Omitted Confounder



Notes: This figure applies the omitted-variable sensitivity analysis of [Cinelli and Hazlett \[2020\]](#) to equation (3) in the 2015–2024 firm-year sample. The panels show changes in the policy coefficient (left) and its t-statistic (right) under alternative confounder strengths, benchmarked to city per capita GDP. The regression uses robot-adoption exposure, the lagged migrant-inclusiveness index, the baseline controls, and firm and year fixed effects. The t-statistic contours use the conventional OLS standard error implemented by `sensemkr`, whereas the baseline table reports standard errors clustered at the two-digit industry level.

C.3 Instrumental-Variable Checks

The next set of tables reports the exclusion-restriction tests, alternative IV estimators, network-placebo exercises, controls for conditions in connected cities, and tests based on pre-policy characteristics and trends. Appendix B.7 describes the purpose and construction of these exercises.

TABLE C7 Dialect-Network IV: Migration and Negative-Control Tests

	(1)	(2)	(3)
Panel A: Direct Migration Channel			
Specification	Direct Reduced Form	+ Policy Index	FWL Residualization
Dependent Variable:	Resident-Registered Population Gap Share		
Dialect-Network IV	0.897	0.633	0.632
<i>p</i> -value			
Observations	1,638	1,638	1,638
Baseline Controls and FE	✓	✓	✓
Panel B: Policy-Pre City-Characteristic Negative-Control Outcomes			
Test	Statistic	<i>p</i> -value	Observations
Six Individual NCOs, Bonferroni Adjusted	–	All $p \geq 0.396$	157–159
Joint NCO Test	$F = 1.489$	0.226	157
Panel C: Future-IV Negative-Control Exposure			
Test	Statistic	<i>p</i> -value	Observations
Linear Specification	$F = 2.020$	0.167	8,296
Linear and Quadratic Terms	$F = 1.289$	0.293	8,296
Quadratic Terms and IV–NCI Interaction	$F = 0.898$	0.455	8,296

Notes: This table examines migration, pre-policy city characteristics, and future policy exposure as potential alternative paths linking the dialect-network instrument to robot exposure. The instrument is defined in equation (4). Panel A uses city-year observations and the resident-registered population gap share as a stock-based migration proxy. Columns (1)–(3) report *p*-values for the instrument in the direct reduced form, after controlling for the local policy index, and after Frisch–Waugh–Lovell residualization, respectively. These specifications include city and year fixed effects and city-level controls. Panel B uses city observations to test six pre-policy characteristics and reports Bonferroni-adjusted *p*-values for the individual tests together with a joint *F*-test. Panel C uses firm-year observations to test future network exposure conditional on realized exposure. The three rows report a linear specification, a specification with linear and quadratic terms, and a specification with quadratic terms and the current-IV–future-IV interaction. These exercises follow the negative-control approach discussed by [Danieli et al. \[2026\]](#); Appendix B.7 describes the implementation. Standard errors are clustered by city in Panel A, by province in Panel B, and by two-digit industry in Panel C. The table reports *p*-values directly. “–” denotes not applicable.

TABLE C8 Dialect-Network IV Estimates Using Fuller Estimators

Estimator:	Industrial Robot Adoption ($Install_{ict}$)		
	(1) 2SLS Baseline	(2) Fuller(1)	(3) Fuller(4)
Panel A: Second-Stage Estimates			
Lagged Migrant-Inclusiveness Index	0.2789*** [0.0947]	0.2787*** [0.0946]	0.2783*** [0.0945]
Panel B: First Stage — Dependent Variable: Lagged Migrant-Inclusiveness Index			
Dialect-Network IV (Z)	0.0034*** [0.0002]	0.0034*** [0.0002]	0.0034*** [0.0002]
Panel C: Reduced Form — Dependent Variable: Industrial Robot Adoption			
Dialect-Network IV (Z)	0.0010*** [0.0003]	0.0010*** [0.0003]	0.0010*** [0.0003]
Panel D: Identification Statistics			
First-Stage F	342.50	342.50	342.50
KP rk Wald F	342.45	342.45	342.45
MOP Effective F	342.45	342.45	342.45
MOP 10% Critical Value	23.11	23.11	23.11
First-Stage Relevance p -value	< 0.001	< 0.001	< 0.001
Endogeneity Test F (p)	3.76 (0.061)	—	—
Wu–Hausman F (p)	3.52 (0.061)	—	—
Conventional t Statistic	2.95	—	—
5% Critical Value	2.03	—	—
95% Confidence Interval	[0.086, 0.471]	—	—
Observations	9,117	9,117	9,117
<i>Fixed Effects:</i>			
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
<i>Controls:</i>			
Firm Controls	✓	✓	✓
City Controls	✓	✓	✓

Notes: This table compares IV estimators for equation (3), using the dialect-network exposure defined in equation (4) to instrument for lagged local migrant inclusiveness. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. Column (1) repeats the baseline 2SLS specification, while columns (2)–(3) use the Fuller(1) and Fuller(4) estimators [Fuller, 1977]. Panel A reports the second-stage policy coefficient for robot-adoption exposure. Panel B reports the first stage for local policy, Panel C reports the reduced form for robot exposure, and Panel D reports identification and inference statistics. KP denotes Kleibergen–Paap, and MOP denotes Montiel Olea–Pflueger. The first-stage and reduced-form entries are repeated because all three estimators use the same instrument, estimation sample, controls, and fixed effects. Table 2 reports LIML estimates and tF -adjusted inference. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. “—” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C9 Dialect-Network IV: Source-Shock Recentering and Network Placebo

	(1)	(2)
Specification:	Recentered IV	Network Placebo
Panel A: Recentered Dialect-Network IV		
Second-Stage Coefficient	0.1786* [0.0906]	–
First-Stage F	358.26	–
KP rk Wald F	358.22	–
Panel B: Randomized-Network Placebo		
Actual First-Stage F	–	342.50
Placebo First-Stage F (Median / 95th / 99th)	–	5.09 / 39.73 / 64.30
Empirical p -value (Actual F vs. Placebo)	–	< 0.001
Observations	9,117	9,117
Firm FE	✓	✓
Year FE	✓	✓
Baseline Controls	✓	✓

Notes: This table examines the construction of the dialect-network instrument by varying source-policy shocks and their assignment to network positions. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The outcome is robot-adoption exposure, and the endogenous regressor is lagged local migrant inclusiveness. Panel A reports the second-stage policy coefficient and first-stage statistics after recentering the source-policy shocks used in the instrument. Panel B randomly reassigns source-city policy identities across network positions and reports the observed first-stage F -statistic, the median and 95th and 99th percentiles of the placebo distribution, and the empirical p -value comparing the observed statistic with that distribution. These exercises examine the respective roles of shocks and exposure weights discussed by Goldsmith-Pinkham et al. [2020] and Borusyak et al. [2022]. KP denotes Kleibergen–Paap. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. “–” denotes not applicable. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C10 Dialect-Network IV: Zero Local Policy Flow and External Economic Conditions

	Firm Outcomes				
Dependent Variable:	(1) Industrial Robot Adoption	(2) Employee Wage per Worker			
Panel A: Zero Local Policy-Flow Observations					
Dialect-Network IV (Z)	-0.0004 [0.0020]	0.2679 [2.3121]			
p -value	0.828	0.909			
Observations	660	660			
Fixed Effects:					
Unit FE	✓	✓			
Year FE	✓	✓			
Controls:					
Baseline Controls	✓	✓			
	Baseline	Additional Network-Weighted Controls			
	(1)	(2)	(3)	(4)	(5)
Specification:	Baseline IV	GDP Growth	Firm Wages	Insurance Coverage Growth	GDP and Insurance Growth
Panel B: Network External-Variable Controls					
Lagged Migrant-Inclusiveness Index	0.2789*** [0.0947]	0.2783** [0.1241]	0.2820*** [0.0944]	0.2281** [0.0863]	0.2334** [0.1115]
p -value	0.006	0.032	0.005	0.012	0.044
First-Stage F	342.50	369.48	337.40	284.72	308.32
Observations	9,117	9,117	9,117	9,117	9,117
Fixed Effects:					
Firm FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Controls:					
Firm Controls	✓	✓	✓	✓	✓
City Controls	✓	✓	✓	✓	✓

Notes: This table examines the relationship between the dialect-network instrument and selected firm outcomes when local policy flow is zero, and whether economic conditions in source cities account for the IV estimates. Panel A restricts the 2015–2024 panel to observations with a zero lagged local policy index and reports reduced-form coefficients on the dialect-network instrument. Columns (1)–(2) use firm-year robot exposure and wages per worker, respectively. Panel A includes the fixed effects indicated in the table. Panel B uses the full firm-year IV sample and reports second-stage coefficients on lagged local policy. Column (1) reports the baseline specification. Columns (2)–(4) add network-weighted GDP growth, listed-firm wages, and unemployment-insurance participation growth, respectively; column (5) adds GDP growth and insurance-participation growth jointly. These external variables enter both stages of the IV specification. Panel B includes the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C11 Dialect-Network Exposure and Pre-Period Economic Conditions

Dependent Variable:	Pre-Trend	Firm Balance			
	(1) Robot Trend 2012–2014	(2) Firm Size 2014	(3) ROA 2014	(4) Leverage 2014	(5) Sales Growth 2014
Panel A: Firm-Level Trends and Baseline Characteristics					
Dialect-Network IV ($Z_{c,2015}$)	-0.1427	0.0934	-0.0052	0.0297	-0.0227
	[0.1121]	[0.0988]	[0.0040]	[0.0213]	[0.0213]
<i>p</i> -value	0.214	0.353	0.201	0.176	0.295
Observations	941	909	909	909	909
Province FE	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓
Panel B: City-Level Pre-Treatment Trajectories					
Dependent Variable:	(1) Per Capita GDP Trend 2012–2014	(2) Science-Expenditure Trend, 2012–2014	(3) Average GDP Growth 2012–2014		
Dialect-Network IV ($Z_{c,2015}$)	-322.2763	255.8190	0.1241		
	[218.5268]	[2686.8526]	[0.3966]		
<i>p</i> -value	0.153	0.925	0.757		
Observations	283	283	283		
Province FE	✓	✓	✓		
Industry FE	–	–	–		

Notes: This table regresses pre-period outcomes on the 2015 dialect-network instrument to examine its relationship with economic conditions preceding the main estimation window. Panel A uses firm observations. Column (1) uses the firm’s robot-exposure trend during 2012–2014; columns (2)–(5) use firm size, return on assets, leverage, and sales growth in 2014. These regressions include province and industry fixed effects. Panel B uses city observations and reports relationships with trends in per-capita GDP and science expenditure and with average GDP growth during 2012–2014. These regressions include province fixed effects. The dependent variables and sample sizes are reported by column. The instrument follows equation (4), and Appendix B.7 describes the pre-period design. Standard errors are shown in brackets, and coefficient *p*-values are reported directly. “–” denotes not applicable.

C.4 Supplementary Heterogeneity and Mechanism Results

The supplementary heterogeneity results extend the main comparisons to task exposure, product-market competition, prior corporate violations, labor expenditure, and overtime. Prior violations are defined from announcements dated between 2010 and 2014 and remain fixed during the estimation period, as detailed in Appendix B.4. The remaining tables report policy-dimension results for the cost outcomes and additional firm and local classifications.

TABLE C12 Migrant Inclusion and Robot Exposure by Tasks, Competition, and Prior Violations

Panel A. Routine Low-Skill Share				
Subgroups	High Share		Low Share	
Lagged Migrant-Inclusiveness Index	0.20* [0.1139]		-0.04 [0.0666]	
Observations	4,480		4,483	
R-squared	0.86		0.89	
Outcome Mean	23.25		22.23	
<i>p</i> -value for Difference			0.004***	
Firm FE	✓		✓	
Year FE	✓		✓	
Baseline Controls	✓		✓	
Panel B. Product-Market Competition				
Subgroups	Lower Industry Boone	Higher Industry Boone	Lower Firm Markup	Higher Firm Markup
Lagged Migrant-Inclusiveness Index	0.07* [0.0326]	0.17*** [0.0380]	0.15* [0.0794]	0.03 [0.0434]
Observations	4,578	4,391	4,356	4,347
R-squared	0.95	0.88	0.87	0.90
Outcome Mean	15.76	30.26	21.60	23.94
<i>p</i> -value for Difference	0.074*		0.080*	
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓
Panel C. Prior Corporate Violations				
Subgroups	Yes		No	
Lagged Migrant-Inclusiveness Index	0.22*** [0.0653]		0.06 [0.0471]	
Observations	2,614		6,503	
R-squared	0.86		0.86	
Outcome Mean	21.84		23.04	
<i>p</i> -value for Difference			0.048**	
Firm FE	✓		✓	
Year FE	✓		✓	
Baseline Controls	✓		✓	

Notes: This table reports split-sample estimates of equation (3) by task exposure, product-market competition, and pre-sample corporate violation records. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure, and the reported coefficient is for the one-year-lagged aggregate policy index. Panel A splits firms at the pre-treatment sample median of the routine low-skill share. Panel B uses pooled-median splits of the fine-industry Boone-type index and the firm cost-markup measure. The Boone index retains the sign of the estimated market-share–average-cost elasticity; lower markups indicate stronger competitive pressure. Panel C distinguishes 272 firms with company-level violations announced during 2010–2014 from 670 firms without such records. Appendix B.4 defines the grouping variables. The reported *p*-values for coefficient differences are obtained from bootstrap tests of equality between the paired estimates. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C13 Migrant Inclusion and Robot Exposure by Labor Expenditure and Overtime

	Labor-Related Expenditure		Annual Overtime	
	(1)	(2)	(3)	(4)
Panel A: Labor Expenditure and Overall Overtime				
Subgroups	High Per-Worker Increase	Low Per-Worker Increase	High	Low
Lagged Migrant-Inclusiveness Index	-0.0553	0.1520**	0.1661***	-0.0195
	[0.0667]	[0.0684]	[0.0556]	[0.0731]
<i>p</i> -value for Difference	< 0.001***		0.004***	
Observations	4,028	4,024	4,492	4,483
R-squared	0.90	0.86	0.89	0.87
Outcome Mean	20.38	22.94	22.08	23.26
Panel B: Regular and Weekend Overtime				
	Regular Overtime		Weekend Overtime	
Subgroups	High	Low	High	Low
Lagged Migrant-Inclusiveness Index	0.1628**	-0.0094	0.1251**	0.0521
	[0.0610]	[0.0617]	[0.0552]	[0.0526]
<i>p</i> -value for Difference	0.004***		0.192	
Observations	4,491	4,487	4,489	4,489
R-squared	0.89	0.87	0.89	0.86
Outcome Mean	21.95	23.48	22.85	22.61
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓

Notes: This table reports split-sample estimates of equation (3) by labor expenditure and overtime intensity. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure, and the reported coefficient is for the one-year-lagged aggregate policy index. In Panel A, columns (1)–(2) compare firms with higher and lower growth in labor-related expenditure per worker, while columns (3)–(4) compare firms with higher and lower annual overtime intensity. In Panel B, columns (1)–(2) use regular overtime, and columns (3)–(4) use weekend overtime. Each pair uses the observations available for the corresponding grouping variable. The reported *p*-values for coefficient differences are obtained from bootstrap tests of equality between the two subgroups; 0.000 reflects rounding. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C14 Policy Dimensions, Employment Costs, Social Protection, and Labor Composition

	Compensation Costs			Social Protection		Labor Composition	
	(1) Accrued Comp.	(2) Cash Comp.	(3) Wage per Worker	(4) Firm Insurance Rate	(5) City Insurance Coverage	(6) Log Low-Skill Workers	(7) Log Graduate Workers
Hukou/Residence	0.48 [0.7599]	4.65 [3.0624]	0.20 [0.2093]	-5.41×10^{-4} [0.0003]	0.75 [0.4789]	-4.29×10^{-3} [0.0026]	5.33×10^{-3} [0.0048]
Social Security	0.85 [1.1099]	5.51* [3.0629]	0.19 [0.2132]	-2.51×10^{-4} [0.0002]	1.42*** [0.3598]	9.50×10^{-4} [0.0018]	2.24×10^{-3} [0.0039]
Public-Service Access	1.73** [0.6654]	6.96*** [2.0493]	0.36* [0.1818]	-1.58×10^{-4} [0.0002]	2.21*** [0.3265]	-3.11×10^{-3} [0.0016]	1.26×10^{-3} [0.0030]
Labor-Rights Protection	0.13 [0.2927]	1.24 [1.0078]	0.16** [0.0790]	-5.47×10^{-5} [0.0001]	1.08*** [0.2159]	-1.02×10^{-3} [0.0009]	3.32×10^{-3} [0.0019]
Employment/ Entrepreneurship	1.06 [0.7587]	5.72** [2.6825]	0.25 [0.1704]	5.01×10^{-6} [0.0002]	-0.14 [0.3750]	-5.20×10^{-4} [0.0015]	2.54×10^{-3} [0.0031]
Administrative Relief	0.18 [0.6136]	-0.84 [2.0230]	0.06 [0.1837]	5.17×10^{-5} [0.0002]	-2.89*** [0.2898]	-1.23×10^{-3} [0.0013]	-2.99×10^{-4} [0.0026]
Observations	9,102	9,117	9,117	9,058	8,871	7,380	8,581
Outcome Mean	158.76	916.08	137.81	0.15	355.51	7.39	4.84
Outcome SD	318.22	1432.06	60.64	0.05	407.78	1.21	1.62
Baseline Controls and FE	✓	✓	✓	✓	✓	✓	✓

Notes: This table reports separate regressions for each policy dimension and outcome, relating policy content to employment costs, formal social protection, and workforce composition. The unit of observation is the firm-year in the 2015–2024 listed-manufacturing panel; city unemployment-insurance participation is matched to firm observations. Each coefficient comes from a regression using one policy subindex lagged by one year and the controls and fixed effects in equation (3). Columns (1)–(3) use accrued compensation and cash compensation paid, both measured in millions of yuan, and wages per worker, measured in thousands of yuan. Columns (4)–(5) use the firm social-insurance rate and city unemployment-insurance participation, measured in units of 10,000 persons. Columns (6)–(7) use the logarithms of low-skill and graduate employment. Appendix B.3 defines the outcomes. Observation counts, means, and standard deviations are reported separately for each outcome. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors in brackets are clustered at the two-digit industry level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C15 Policy-Dimension Heterogeneity: Tests of Equality across Firm Groups

Bootstrap p -value for Equality of Paired-Group Coefficients					
Panel A: Labor-Cost and Workforce Exposure					
Policy Dimension	Cost Exposure			Workforce / Expenditure	
	Labor Intensity	FCP Constraint	Prior Violations	Routine Low-Skill	Labor Expenditure
Hukou/Residence	0.306	0.156	0.314	0.064	0.154
Social Security	0.126	0.122	0.048	0.012	0.004
Public-Service Access	0.022	0.060	0.146	< 0.001	< 0.001
Labor-Rights Protection	0.056	0.030	0.034	< 0.001	< 0.001
Employment/Entrepreneurship	0.180	0.150	0.062	0.014	0.034
Administrative Relief	0.300	0.164	0.304	0.014	0.002
Panel B: Administrative Reach and Overtime Intensity					
Policy Dimension	Administrative Reach		Overtime Intensity		
	Political Connection	Government Distance	Annual	Regular	Weekend
Hukou/Residence	0.186	0.178	0.056	0.054	0.012
Social Security	0.400	0.124	0.146	0.488	0.244
Public-Service Access	0.120	0.400	0.026	0.218	0.196
Labor-Rights Protection	0.330	0.296	0.076	0.112	0.300
Employment/Entrepreneurship	0.242	0.330	0.056	0.176	0.118
Administrative Relief	0.108	0.446	0.352	0.224	0.302

Notes: This table reports bootstrap p -values for tests of equality between policy-dimension coefficients across the two groups indicated by each column. Each cell is based on separate regressions of firm robot-adoption exposure on the corresponding one-year-lagged policy subindex, following equation (3), in the 2015–2024 listed-manufacturing sample. Panel A covers labor intensity, the mean FCP score during 2010–2014, prior company violations, the routine low-skill share, and growth in labor-related expenditure per worker. Higher FCP values indicate tighter financial constraints; the violation split distinguishes firms with and without company-level violations announced during 2010–2014. Panel B covers political connections, distance to government, and annual, regular, and weekend overtime. Distance is measured in kilometers. Appendix B.4 defines the grouping variables. All underlying regressions include the baseline firm and city controls and firm and year fixed effects. Entries are p -values rather than policy coefficients. Bold entries indicate rejection of coefficient equality at the 5% level.

TABLE C16 Policy-Dimension Estimates by Legal Environment and Governance Responsiveness

Subgroups	Local Legal Environment		Shareholder-Meeting Frequency	
	(1) Lower	(2) Higher	(3) Low	(4) High
Hukou/Residence	0.0352 [0.1941]	-0.0043 [0.2083]	0.1986 [0.2639]	-0.0570 [0.1309]
<i>p</i> -value for Difference	0.426		0.186	
Social Security	0.0471 [0.1356]	0.6911** [0.2961]	0.0457 [0.2261]	0.4441 [0.3446]
<i>p</i> -value for Difference	0.004***		0.058*	
Public-Service Access	-0.0881 [0.0883]	0.7941*** [0.2676]	0.2410* [0.1353]	0.4928* [0.2623]
<i>p</i> -value for Difference	< 0.001***		0.072*	
Labor-Rights Protection	0.0227 [0.0450]	0.2309** [0.0880]	-0.0111 [0.0663]	0.2176* [0.1132]
<i>p</i> -value for Difference	0.018**		0.012**	
Employment/Entrepreneurship	0.0607 [0.1186]	0.4288** [0.2060]	-0.1117 [0.1164]	0.2002 [0.1829]
<i>p</i> -value for Difference	0.038**		0.076*	
Administrative Relief	-0.0065 [0.1068]	-0.3629* [0.1918]	-0.1186 [0.1021]	-0.5627** [0.2193]
<i>p</i> -value for Difference	0.022**		0.006***	
Observations	4,779	4,256	3,318	5,633
R-squared	0.91	0.88	0.88	0.87
Outcome Mean	21.35	24.32	21.37	23.43
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓

Notes: This table reports split-sample policy-dimension regressions by provincial legal and market institutions and firm governance responsiveness. The unit of observation is the firm-year, and the sample covers Chinese listed manufacturing firms during 2015–2024. The dependent variable is robot-adoption exposure. Each policy-dimension coefficient is estimated separately using the one-year-lagged subindex and the specification in equation (3). Columns (1)–(2) compare lower and higher values of the provincial index of market-intermediary development and legal institutions. Columns (3)–(4) compare firms with lower and higher shareholder-meeting frequency, which proxies for governance responsiveness. For each policy dimension, the reported *p*-value tests equality between the paired subgroup coefficients; 0.000 reflects rounding. All regressions include the baseline firm and city controls and firm and year fixed effects. Standard errors are clustered at the two-digit industry level and shown in brackets. Each column contains six regressions, and R-squared values are reported to two decimal places. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

TABLE C17 Policy-Dimension Estimates by Environmental Pressure and Market Competition

Subgroups	Environmental Keywords		Environmental Penalties		Industry Boone		Firm Markup	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lower	Higher	Lower	Higher	Lower	Higher	Lower	Higher
Hukou/Residence	-0.12 [0.26]	0.15 [0.12]	-0.12 [0.19]	0.28* [0.14]	-0.03 [0.08]	-0.03 [0.25]	-0.02 [0.12]	0.10 [0.25]
<i>p</i> -value for Difference	0.150		0.078*		0.490		0.328	
Social Security	0.21 [0.23]	0.36** [0.14]	-0.10 [0.07]	0.42** [0.19]	0.02 [0.09]	0.45*** [0.15]	0.28* [0.14]	0.13 [0.10]
<i>p</i> -value for Difference	0.274		0.026**		0.038**		0.262	
Public-Service Access	0.28** [0.13]	0.62*** [0.20]	-0.16 [0.14]	0.59*** [0.19]	0.06 [0.06]	0.59*** [0.15]	0.47** [0.19]	0.13 [0.10]
<i>p</i> -value for Difference	0.032**		< 0.001***		< 0.001***		0.034**	
Labor-Rights Protection	0.11 [0.11]	0.02 [0.07]	-0.01 [0.06]	0.24*** [0.08]	0.06 [0.04]	0.16*** [0.05]	0.15* [0.08]	0.03 [0.05]
<i>p</i> -value for Difference	0.164		0.006***		0.096*		0.112	
Employment/ Entrepreneurship	-0.17 [0.13]	0.12 [0.08]	-0.29** [0.12]	0.08 [0.07]	0.01 [0.09]	0.17 [0.18]	0.08 [0.10]	0.04 [0.09]
<i>p</i> -value for Difference	0.068*		0.036**		0.212		0.418	
Administrative Relief	-0.18 [0.15]	-0.51*** [0.16]	-0.05 [0.16]	-0.53* [0.28]	-0.10* [0.05]	-0.49*** [0.14]	-0.32*** [0.11]	-0.22 [0.14]
<i>p</i> -value for Difference	0.024**		0.002***		0.002***		0.300	
Observations	4558	4426	4448	4519	4578	4391	4356	4347
R-squared	0.89	0.87	0.87	0.89	0.95	0.88	0.87	0.90
Outcome Mean	25.40	19.87	18.36	26.99	15.76	30.26	21.60	23.94
Firm FE	✓	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Baseline Controls	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table reports split-sample policy-dimension estimates by environmental-policy pressure and product-market competition. The unit of observation is the firm-year in the 2015–2024 panel of Chinese listed manufacturing firms. The dependent variable is robot-adoption exposure. Each coefficient comes from a separate regression using the indicated policy subindex lagged by one year, following equation (3). Columns (1)–(2) compare lower and higher values of the environmental-policy keyword measure, while columns (3)–(4) compare lower and higher environmental penalties. These variables capture environmental-policy language and realized enforcement, respectively. Columns (5)–(6) use the signed fine-industry Boone-type index, and columns (7)–(8) use the firm cost-markup measure; both variables are split at their pooled firm-year medians. Lower markups indicate stronger competitive pressure. For each pair of columns, the reported *p*-value tests equality between the subgroup coefficients; 0.000 reflects rounding. All regressions include the baseline controls and firm and year fixed effects. Standard errors are clustered at the two-digit industry level and shown in brackets beneath the coefficients. Each column contains six regressions; the reported R-squared is the common value rounded to two decimals, or a range when the rounded values differ. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

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